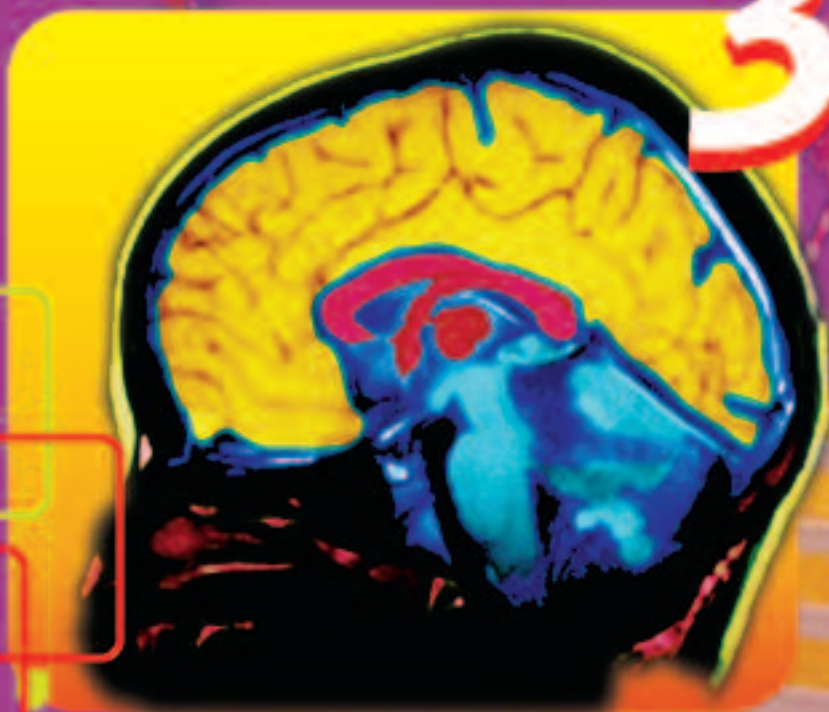


Practical Experiments in **BIOLOGY**

3



Satya Naidu



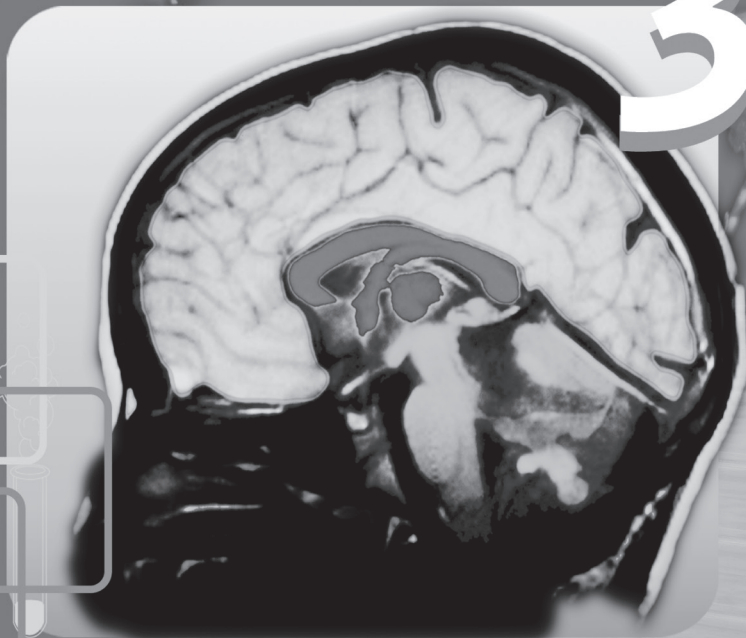
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Practical Experiments in

BIOLOGY

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Satya Naidu

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I wish to thank Carol Andrews for reviewing the original drafts and Pam Robinson for her suggestions and for providing materials for the experiments.

K. Satya Naidu

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INTRODUCTION

Practical Experiments in Biology Book 3 is part of a four-book series that presents experiments from different areas of study that are relevant to senior Biology students. Book 3 focuses on homeostasis, plant tropisms, the nervous system, animal behaviour, cell division and microbiology. Teacher's notes and useful websites for each experiment are provided at the start of the book. These include strategies for enhancing student understanding of the experiment and suggestions for overcoming difficulties. Internet addresses for relevant websites are also provided to encourage students to undertake further research.

The experiments are presented in a self-contained format that Biology students will find easy to follow. Each experiment includes a summary of relevant background knowledge and is structured according to aim, hypothesis, equipment, safety rules, method, results, analysis of results and conclusion. Tables and space for diagrams are provided for students to record their results. The clear, step-by-step structure will encourage the efficient use of experimental time and give guidance to students on how to write scientific reports. Suggested answers and expected results have also been provided where appropriate.

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TEACHER'S NOTES AND USEFUL WEBSITES

Experiment 1: Homeostatic control of temperature after exercise

Suggestions for teachers

- Prior to the experiment, explain the role of the negative feedback system in controlling temperature, pulse rate and CO₂.
- Show students how to measure pulse rates and temperature.
- Make sure the students take readings immediately after the exercise.
- Tell the students to be careful in dropping the NaOH solution into water to make sure they count accurately. Remind them that it is important to have a control in order to compare the colour change.
- Make the students aware of appropriate websites.

Websites

<http://www.campbell.edu/faculty/bergemann/scott2.html>
<http://www.uic.edu/depts/bioe/Bioproject/Ban,%20T-%20Body%20Temperature%20Control.doc>
http://www3.fhs.usyd.edu.au/bio/homeostasis/Temp_Control_System.htm
<http://en.wikipedia.org/wiki/Homeostasis>
<http://fog.ccsf.cc.ca.us/~mmalacho/physio/oll/Lesson1/hstasis.html#toc2>
http://www.nyp.org/health/cd_rom_content/adult/cardiac/vital.htm

Experiment 2: Phototropism and geotropism in plants

Suggestions for teachers

- Discuss phototropism, geotropism and stimulus response models with the students before the experiment. Show relevant videos and images if they are available.
- If possible, provide petri dishes perforated with holes for part C.
- Make students aware of appropriate websites.

Websites

<http://school.discovery.com/lessonplans/programs/tropisms/>
http://koning.ecsu.ctstateu.edu/Plant_Physiology/phototropism.html
<http://www.biochem.arizona.edu/classes/bioc462/462bH2002/462bHonorsProjects/462bHonors2001/Schmitz/planthormones.htm>
http://www.fofweb.com/Onfiles/SEOF/Science_Experiments/4-26.pdf
http://eosweb.larc.nasa.gov/EDDOCS/Wavelengths_for_Colors.html#blue

Experiment 3: Investigating reflex actions

Suggestions for teachers

- Students must have a good understanding of the central nervous system and reflex actions before doing this experiment.
- Students should know the difference between voluntary and involuntary actions.
- Make the students aware of appropriate websites.

Websites

http://www.rgi.tut.fi/kurssit/71221/nerve/nerve_velocity.pdf
<http://faculty.washington.edu/chudler/chreflex.html>
<http://www.ktca.org/newtons/13/rlexes.html>
<http://www.neuro.wustl.edu/neuromuscular/mother/reflex.html>
<http://www.explorelearning.com/index.cfm?method=cResource.dspDetail&ResourceID=43->
<http://www.gen.umn.edu/courses/1135/lab/reflexlab/reflexlab.html>
<http://www.pinkmonkey.com/studyguides/subjects/biology-edited/chap23/b2323401.asp>

Experiment 4: Sense receptors in the skin

Suggestions for teachers

- The students must have a good background knowledge of the nervous system and sense organs before doing this experiment.
- When testing water temperature, make sure the water is not boiling hot.
- When doing the two-point discrimination activity, emphasise that the two tips of the toothpicks should be gently applied to the skin and that both tips should touch the skin at the same time.
- Make the students aware of appropriate websites.

Websites

<http://www.enchantedlearning.com/subjects/anatomy/skin/label/labelanswers.shtml>
<http://faculty.washington.edu/chudler/chtouch.html>
<http://faculty.washington.edu/chudler/pdf/mmtg.pdf>
<http://www.alab.t.u-tokyo.ac.jp/~shino/research/pdf/1998CGA.pdf>

Experiment 5: Orientation movements exhibited by planaria

Suggestions for teachers

- Keep the lights in the room low during this experiment, as stray light can cause unexpected results.
- Keep the planaria in pond water or aged tap water.
- Planaria can be collected from a local pond, but several species are available from biological supply companies.
- Feed the planaria every few days. A small amount of canned cat food is good, but change the water immediately after feeding.
- Make the students aware of appropriate websites.

Websites

<http://psychology.okstate.edu/faculty/abramson/plan.html>
http://ebiomed.com/gall/classics/Plan/plan_about.html
http://www.rowan.edu/mars/depts/biology/faculty/richmond/Invertebrate_zoology_2003/Labs/Planaria_etc.doc
<http://fire.biol.wvu.edu/hooper/204lab07animals1.pdf>

Experiment 6: Social and aggressive behaviour in fish

Suggestions for teachers

- Students should have background knowledge of the advantages and disadvantages of social behaviour and aggressive behaviour in fish.

- The students should observe the behaviour of the fish carefully and record only what they observe. They should not interpret the behaviour in human terms (emotions such as anger, happiness, sad).
- Make sure the students do not put two male Siamese fighting fish in the same jar because they can kill each other.
- Use warm water at 28°C for catfish and Siamese fighting fish. Maintain the pH at 7.
- Make the students aware of appropriate websites.

Websites

<http://www.marietta.edu/~biol/introlab/schooling.pdf>
http://www.planetcatfish.com/catelog/help_anatomy.htm
http://vcs.abdn.ac.uk/BIO_SOIL/trout/intro.html - interactive videos
<http://www.science.mcmaster.ca/psychology/psych3s3sbe/practical5/outline.pdf>
<http://home.comcast.net/~nathanconnell/Bio103Term.html>
<http://aci.mta.ca/Courses/Biology/3401/Labs/lab4.pdf>
http://www.zi.ku.dk/popecol/Kurser/Blok2_03/Oevelsesmateriale/oevlse2_u51_Slides.pdf
http://www.nature.com/nsu/nsu_pf/020916/020916-14.html
<http://freshaquarium.about.com/cs/anabantids2/p/betta.htm>

Experiment 7: Modelling mitosis

Suggestions for teachers

- Students must have a good understanding of how and where mitosis takes place.
- Show models and a video on mitosis to the class before the experiment.
- Make the students aware of appropriate websites.

Websites

<http://www.accessexcellence.org/AB/GG/mitosis.html>
http://www.wooster.edu/biology/wmorgan/bio306/Modeling_Mitosis.html
<http://www-Saps.plantsci.cam.ac.uk/worksheets/ssheets/ssheet17.htm>
http://www.wooster.edu/biology/wmorgan/bio306/Observing_Plant_Mitosis.html
<http://www.lima.ohio-state.edu/biology/mitosis.html>
<http://tidepool.st.usm.edu/crswr/roottip.html>

Experiment 8: Modelling meiosis

Suggestions for teachers

- Students should have a good understanding of how meiosis differs from mitosis.
- Students should complete the modelling of mitosis (experiment 7) before the modelling of meiosis.
- Provide charts, models and core textbooks for reference.
- Show a video and provide microscope slides of different stages in meiosis and mitosis.
- Make the students aware of appropriate websites.

Websites

<http://www.accessexcellence.org/AB/GG/meiosis.html>
http://www.biosci.uga.edu/almanac/bio_103/notes/apr_3.html
<http://naturalsciences.sdsu.edu/classes/lab2.5/lab2.5.html#anchor29708088>

<http://faculty.clintoncc.suny.edu/faculty/Michael.Gregory/files/Bio%20100/Bio%20100%20online%20laboratory%20exercises/Mitosis/mitosis.htm>

<http://www.wooster.edu/biology/wmorgan/bio306/ModelingMeiosis.html>

<http://faculty.clintoncc.suny.edu/faculty/Michael.Gregory/files/Bio%20101/Bio%20101%20Lectures/Meiosis/meiosis.htm>

<http://www.accessexcellence.org/AB/GG/meiosis.html>

http://www.biology.arizona.edu/cell_bio/tutorials/meiosis/07d.html

Experiment 9: Antibiotic resistance in bacteria

Suggestions for teachers

- Clarify the meanings of antibiotics, sensitive, resistant, agar, Kirby-Bauer Method and zone of inhibition.
- Explain lab safety precautions to minimise the possibility of accidents and reinforce aseptic techniques.
- Store the antibiotic discs at –20°C. Keep the container with discs at room temperature for 30 minutes after its removal from the refrigerator. This will prevent condensation forming on the discs, which can make some antibiotics inactivate. Do not use expired discs.
- Make the students aware of appropriate websites.

Websites

www.med.unsw.edu.au/pathology-cds/NL_2.rtf
http://www.woodrow.org/teachers/bi/2000/Antibiotic_Resistance/student_guide___testing_for_ant.html
http://www.biotechnology.gov.au/biotechnologyOnline/PDF/whatis/WS_2.1.1.pdf
<http://www.innvista.com/health/microbes/bacteria/staph.htm>
http://www.cdc.gov/ncidod/dbmd/diseaseinfo/escherichiacoli_g.htm
<http://web.ukonline.co.uk/webwise/spinneret/edexcel/pracw.htm>
<http://www.kent.k12.wa.us/staff/kloschky/MoneransFolder/shapes.html>
<http://people.ku.edu/~jbrown/ecoli.html>

Experiment 10: Human parasites

Suggestions for teachers

- Students should have a good background knowledge of life cycles and adaptations of endoparasites and ectoparasites.
- Make sure that the students know how to use a light microscope and a stereo microscope before they do this experiment.
- Provide the students with charts, videos and reference materials on parasites.
- Make the students aware of appropriate websites.

Websites

<http://cal.vet.upenn.edu/paraav/labs/tapediag.htm>
http://www.biosci.ohio-state.edu/~parasite/labeled_tenia.html
<http://www.scientific-art.com/portfolio%20medicine%20pages/tapeworm.htm>
<http://www.wbvc.bc.ca/external.htm>
<http://www.hsph.harvard.edu/headlice.html> #what
<http://www.ent.iastate.edu/imagegal/phtiraptera/pedhead-labelled.html>
http://cipm.ncsu.edu/ent/Southern_Region/RIPM/CHAP8/LICE/lice_main.htm

HOMEOSTATIC CONTROL OF TEMPERATURE AFTER EXERCISE



Background

Timing: 2 × 45 minute lessons

Thermoregulation is the ability of an organism to keep its body temperature within certain boundaries, even when the temperature of the surroundings is very different. This process is known as **homeostasis**. Homeostasis is the ability of an organism to maintain a stable internal environment irrespective of changes in either the external or internal environment. In animals, homeostasis is controlled by hormonal and nervous systems.

Negative feedback mechanisms tend to produce stability. They act to restore the original state of the organism. They are **stimulus-response** mechanisms, where the response produced affects the stimulus. Humans are **homeotherms** and maintain body temperature in the range 36.1°C–37.8°C. During exercise, the body produces more heat, carbon dioxide and wastes and the demand for oxygen and glucose increases. As the body temperature increases, the temperature receptors in the **skin** and the **hypothalamus** detect the changes in temperature. The **thermoregulatory centre** in the hypothalamus sends electrical impulses to the **effectors** via motor nerves. The blood vessels near the skin dilate and heat is lost. The sweat glands produce more sweat and heat is lost when the sweat evaporates from the skin. The response is a decrease in temperature until normal body temperature is regained. This response affects the original stimulus by the negative feedback system. The heart rate and breathing rate also increase to provide more oxygen and glucose to the exercising muscles. The percentage increase in the heart rate is influenced by a person's age, physical fitness, gender, size and emotions.

Aim

- To find out the effect of moderate exercise on body temperature, carbon dioxide output and pulse rate.
- To infer how negative feedback mechanisms control temperature.

Equipment

- clinical thermometer
- cobalt chloride papers (blue)
- adhesive tape
- 2 conical flasks
- 6 drinking straws
- a rubber band
- distilled water
- 100 mL measuring cylinder
- 0.4% sodium hydroxide solution
- stop watch
- paper towel
- 0.5% phenolphthalein indicator
- rubbing alcohol

SAFETY RULES

- ⚠ Be careful when placing the clinical thermometer under the tongue, because the mercury inside the thermometer is poisonous. Practise reading the thermometer before the experiment.
- ⚠ Be careful when using alkalis and acids. If they spill on your hands, wash your hands with water immediately.
- ⚠ Wash the conical flasks and drinking straws thoroughly after each reading to eliminate experimental errors.
- ⚠ After the experiment, dispose of the solutions in specified containers. Do not pour them into the sink.

Taking measurements

- **Pulse rate:** The pulse can be found on the side of the lower neck, on the inside of the elbow, or at the wrist. Measure your pulse in the wrist in the **radial artery** on the thumb-side of the wrist. Place your index and middle fingers lightly on your wrist until you feel your pulse, or heart beat. Do not use your thumb because it has a strong pulse of its own. Count the number of pulses in 15 seconds and then multiply by 4. This gives your resting pulse rate. **Note:** The normal pulse rate for men is 72 bpm (beats per minute) and for women it is 80 bpm.

The **carotid pulse** on the side of the neck can be taken by placing the first two fingers on the side of the Adam's apple, in the groove in front of the muscle running vertically down the neck.

- **Temperature:** Clean the thermometer with cool, soapy water or rubbing alcohol. Gripping the end opposite the bulb (or furthest from 0°), shake the thermometer until it reads **37°C** or less. Place the thermometer under the tongue and close the mouth, using the lips to hold the thermometer firmly. Leave the thermometer in the mouth for 2 minutes. Read the thermometer by gripping the end opposite the 0° so that the numbers are facing you. Roll the thermometer back and forth between your fingers until you see a silver or red reflection in the column. Compare the end of the column with the degree markings in the lines on the thermometer. Normal body temperature ranges are **36.1°C–37.8°C**. The average body temperature is about **37°C**.
- **Sweating:** Tape a piece of **cobalt chloride paper** on the forehead of the subject. The paper is blue when dry but turns **pink** when wet. Note the time taken for the blue colour to change to pink.
- **Breathing rate:** This is the number of times air is drawn into the lungs per minute. It can be measured by placing your finger under the subject's nose so that you can feel the breath. Count the number of breaths per minute. Repeat for another minute and record the average reading.

Note: The normal respiration rates for a person at rest range from 15 to 20 breaths per minute. After exercise, the breathing rate will range from 60 to 80 breaths per minute. If you have a **respirometer** use it to measure the respiratory rate volume of the breath.
- **CO₂ output:** This is measured by **titration**. CO₂ dissolves in water, forming **carbonic acid**. Measuring how much NaOH is needed to neutralise this acid provides an indication of CO₂ output. If the amount of NaOH needed increases, then the amount of CO₂ released by the body has increased. **Phenolphthalein indicator** is **colourless** in acid and changes to **pink** in alkali.

Method

At rest

- 1 Work in groups of three. One person is the **experimenter** who takes the readings. The second person is the **subject** whose temperature, pulse, breathing, sweat and CO₂ output are to be measured. The third person is the **recorder** who records the results.
- 2 Ask the subject to sit down quietly. Measure their temperature with a clinical thermometer placed under the tongue for 1 minute. At the same time, take the pulse rate by placing two fingers on the wrist or at the neck. Measure and record the breathing rate.
- 3 Place cobalt chloride paper on the subject's forehead. Note any changes in the colour of the cobalt chloride paper, which indicates the amount of sweating. Also note the skin colour.
- 4 Cover a bench with paper towel and place two conical flasks on it. Pour 10 mL of distilled water into one flask. Add 5 drops of the **phenolphthalein** indicator to the water and swirl the flask. Note the colour of the water.
- 5 Use a rubber band to fasten 6 drinking straws together. Ask the subject to blow air through the straws into the water in the conical flask for 30 seconds.
- 6 Add the **sodium hydroxide** solution drop by drop until the colour of the solution changes and remains **pink** for 30 seconds. Use the second conical flask as a control for comparing the colour.

Note: Make sure the subject breathes in through the nose and breathes out through the mouth into the straws.
- 7 Rinse the straws and conical flask thoroughly after each titration.
- 8 After 2 minutes, repeat all the measurements. Record your results in Table 1.

Exercise

- 9 Ask the subject to run for 10 minutes.

After exercise

- 10 As soon as the exercise is completed, ask the subject to sit down. Measure the temperature, pulse rate, breathing rate and NaOH drops required, and note changes in the cobalt chloride paper. Also note the skin colour.
- 11 Repeat the measurements at 2 minute intervals (2, 4, 6 and 8 minutes) after exercise. Record the results in Table 1.

Homeostatic control of temperature after exercise 1.3

- 12 Clean your equipment and pack up.
- 13 Collate the class results in Table 2. In Table 3, calculate the percentage increase in heart rates for each group and the mean.
- 14 Draw three line graphs to show the changes in the measurements before and after exercise. Draw one for temperature, one for pulse rate and one for CO₂ output.

Results

Table 1 Group individual results

	Rest		Exercise	Recovery				
Time (minutes)	0	2		0	2	4	6	8
Pulse rate per minute								
Breathing rate per minute								
Carbon dioxide output (drops of NaOH)								
Temperature (°C)								

- Percentage increase in heart rate
(rate after exercise – rate at rest) ÷ rate at rest × 100 = _____ %
- Change in cobalt chloride paper
Before exercise _____ After exercise _____
- Skin colour
Before exercise _____ After exercise _____

Table 2 Class results

(B = before exercise; A = after exercise)

	Group 1		Group 2		Group 3		Group 4		Group 5		Group 6		Average	
	B	A	B	A	B	A	B	A	B	A	B	A	B	A
Pulse rate per minute														
Breathing rate per minute														
Carbon dioxide output (drops of NaOH)														
Temperature (°C)														

Table 3 Class results: percentage increase in heart rate

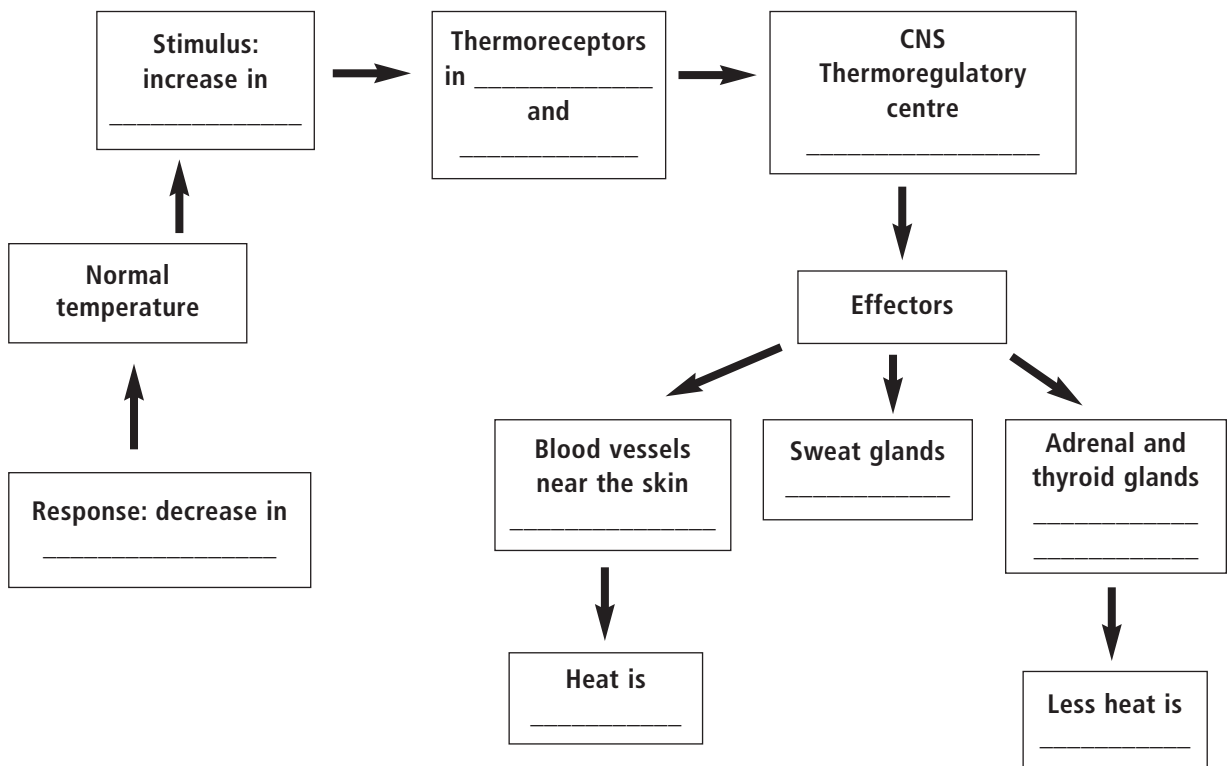
Group	Heart rate at rest	Heart rate after exercise	Increase in heart rate (%)
1			
2			
3			
4			
5			
6			
Mean			

Analysis of results

- 1 What happened to the subject's temperature immediately after exercise?
- 2 Did the temperature change drastically or stay relatively stable with minor fluctuations?
- 3 What happened to the pulse rate immediately after exercise?
- 4 Explain the difference in pulse rate after exercise.
- 5 What happened to the breathing rate after exercise?
- 6 Why did the heart rate and the breathing rate change at the same time after exercise?
- 7 Which gases were being removed and supplied to the leg muscles?
- 8 What other substance does the blood carry to the working muscles to supply energy?
- 9 What substances need to be removed from the muscles by the blood?
- 10 Did the percentage increase in heart rate vary between the 6 groups?
- 11 What factors apart from exercise would affect the rate of heart beat?
- 12 Did the subject sweat a lot after exercise? Explain the purpose of sweating.
- 13 Did you notice any changes in the skin colour of the subject?
- 14 Why do blood vessels near the skin dilate after exercise?
- 15 Use a reference book to find out which part of the brain detects changes in temperature, and which part detects changes in pH or CO₂ in the blood.
- 16 How could you improve the procedure to eliminate experimental errors?

Conclusion

- 1 Complete the diagram. Then explain how negative feedback mechanisms control body temperature.



PHOTOTROPISM AND GEOTROPISM IN PLANTS



Timing: 2 × 45 minute lessons, 1 week observation

Background

Plants adjust to the conditions of their environment through growth responses. Growth towards a source of light is called **positive phototropism**. Growth away from the source of light is called **negative phototropism**. The tips of shoots are positively phototropic. Roots are negatively phototropic. The growth of roots downwards according to gravity is known as **geotropism** or **gravitropism**.

Plant hormones are small organic compounds that influence, at very low concentrations, physiological responses to environmental stimuli. The most important hormone produced by plants is **auxin** or **indole-3-acetic acid (IAA)**. No matter what the concentration, IAA promotes growth in the shoot area of a plant (though higher concentrations promote more growth). At the root tip, high concentrations of auxin **inhibit** growth in the root while small amounts **promote** growth. Auxins regulate **phototropism** and **geotropism** or **gravitropism** and promote stem growth by stimulating **cell elongation**. Auxin is produced at the tip of the shoot and then moves down the shoot, causing the cells in the stem to enlarge. When one side of a plant is darker than the other, more of the auxin goes to the darker side and it then grows faster, bending the stem toward the light. Unequal distribution of auxin is the key to the negative phototropism and positive gravitropism of roots. When a root is placed on its side, **amyloplasts** in the root's tips (organelles containing starch grains) settle by gravity to the bottom of the cells in the root tip and inhibit root cell elongation. So the cells at the top surface of the root elongate, causing the root to grow down.

SAFETY RULES

- ⚠ Wear gloves and a lab coat while doing the experiment.
- ⚠ Keep the working area clean and dispose of the materials in specified containers.
- ⚠ After the experiment, make sure all the equipment is cleaned, dried and put away.
- ⚠ If you are making air holes in the petri dish, wear hot gloves and work in a well ventilated area.

Part A – Phototropism in wheat seedlings grown in white light

Aim To investigate how wheat seedlings respond to light when their coleoptiles are covered with aluminium foil caps and collars.

Hypothesis

Write a hypothesis predicting whether seedlings covered with aluminium foil caps and aluminium foil collars exhibit phototropism.

- Equipment**
- | | | |
|-------------------------|--|----------------------|
| • a bottle with a lid | • 3 aluminium foil collars | • 3 wooden blocks |
| • 50 wheat seeds | • 3 polystyrene cups | • incandescent light |
| • 2% bleach solution | • painted black with slits on one side | • spray bottle |
| • 3 petri dishes | • cotton wool | • plastic ruler |
| • 3 aluminium foil caps | | • scissors |

Method

- 1 One week before the experiment, soak 50 wheat seeds in a bottle of water for 24 hours. Drain off the excess water, put the lid on the bottle and leave the jar in a dark cupboard for 2 days.
- 2 After 2 days, soak the seeds in 2% **bleach solution** for 30 minutes to prevent the growth of mould.
- 3 Select 24 seeds that have **radicles** (primary roots) and **coleoptiles** (a sheath that surrounds the part of the shoot that grows into a leaf). The coleoptiles should be **2–3 cm high**. If they are more than 4 cm high, they are not suitable for this experiment.
- 4 Take 3 clean petri dishes and place cotton wool in each one. Use a spray bottle to wet the cotton wool. Place 6 seedlings in each dish so that the coleoptiles are facing upwards.
- 5 Set up the petri dishes as shown in Figure 2. Label them with your group name and setup A, B and C.

- 6 **Set up A:** Use fine scissors to cut off the top 0.5 cm of the coleoptiles of 3 seedlings. Leave the other 3 seedlings intact.
- 7 **Set up B:** Place aluminium foil caps on 3 seedlings. Leave the other 3 seedlings without caps.
- 8 **Set up C:** Place aluminium foil collars on 3 seedlings. Leave the other 3 seedlings intact without collars. Make sure the tips of the coleoptiles are not covered by the aluminum foil collars.
- 9 Cut a slit into each of three cups, level with the tips of the coleoptiles. Place one cup over each petri dish. Place the incandescent lamp 20 cm from the cups, at the same level as the slit. Use wooden blocks to raise the petri dish height if needed.
- 10 Leave this set up for 3 hours. After 3 hours, remove the cups and observe what has happened to the treated and the untreated coleoptiles in each dish. Record your observations in Table 1.

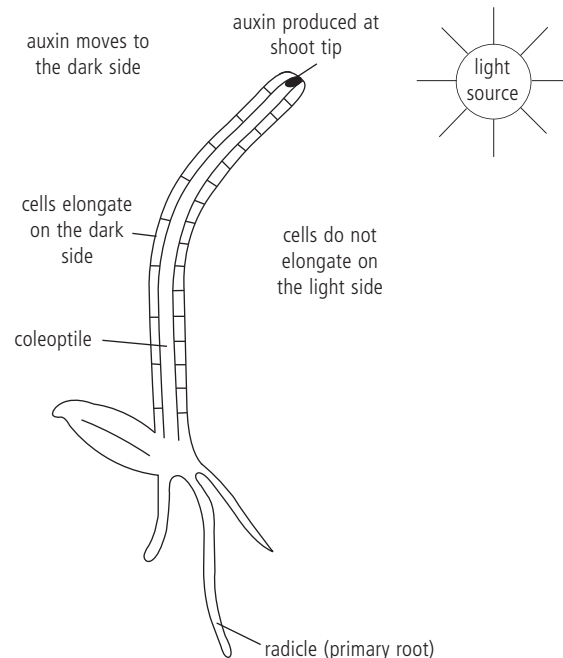


Figure 1: Phototropic response of a plant

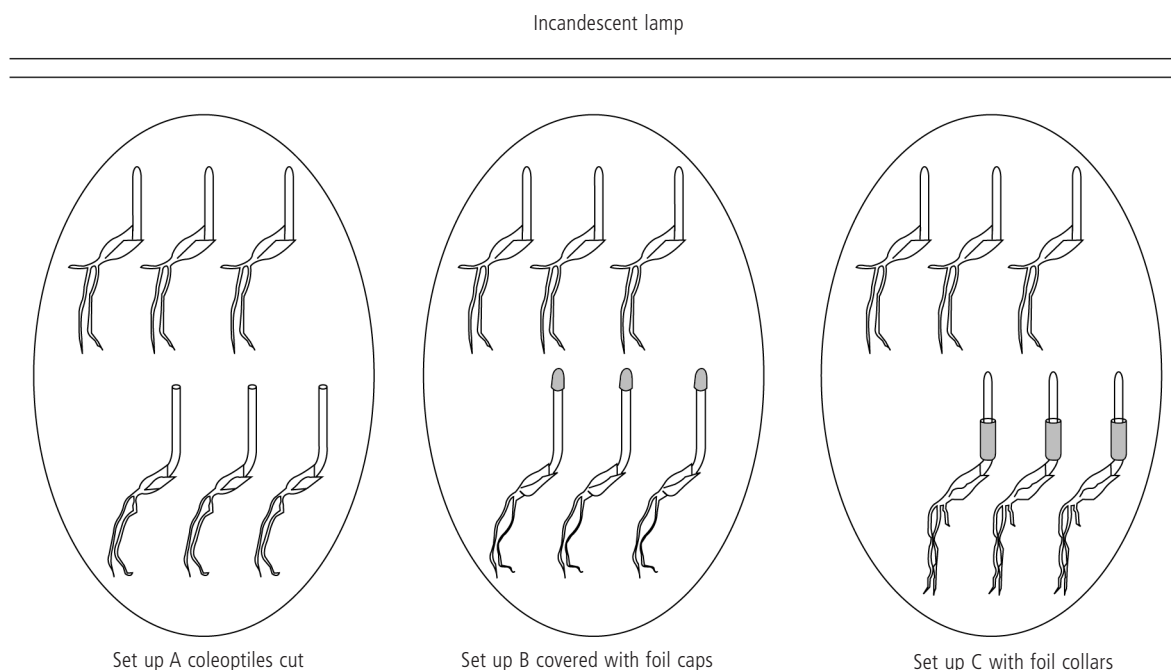


Figure 2: Phototropic responses in seedlings

Results

Table 1

Experimental set up	Response to light	
	Treated seedlings	Untreated seedlings
A: coleoptile tips cut		
B: coleoptile tips covered with caps		
C: coleoptiles with collars		

Analysis of results

- 1 Which seedlings were used as controls in the petri dishes?
- 2 Explain why some seedlings bent towards the light.
- 3 Explain what type of tropism was exhibited by these seedlings.
- 4 Did your findings support your hypothesis?
- 5 What is the advantage of this response in plants?

Part B – Phototropism in wheat seedlings grown under different wavelengths of light

Aim To investigate how plant seedlings respond when exposed to different coloured lights with different wavelengths.

Hypothesis

Write a hypothesis to predict under which coloured light the seedlings will bend more towards light.

Equipment

- Red, blue, green, yellow and clear filters or cellophane
- 5 lightproof cardboard boxes
- 5 pieces of stiff cardboard measuring 3 cm × 3 cm
- sharp knife
- protractor
- spray bottle with water
- 5 incandescent lamps
- 30 germinated wheat seedlings

Method

- 1 Use a knife to cut a rectangular window measuring 3 cm × 2 cm on one side of the cardboard box. Cover the window with two layers of clear cellophane paper.
- 2 Repeat this procedure with 4 other boxes. Cover each slit with a different colour filter or cellophane.
- 3 Under each box, place a petri dish containing 6 germinated wheat seedlings. Place each lamp 40 cm from the slit of one of the five boxes. The coleoptile tips must be level with the slits. Blocks of wood can be placed under the petri dishes to raise their height if needed.
- 4 Place a piece of cardboard on one side of a seedling. Mark the position of the coleoptile before it is exposed to light.
- 5 Place the boxes in a warm, dark place for 4 days. Check the boxes daily and water the seedlings.
- 6 After 24 hours, mark on the cardboard the new position of the coleoptile after phototropic response.
- 7 Measure the angle of bending with a protractor. Record your measurements in Table 2.

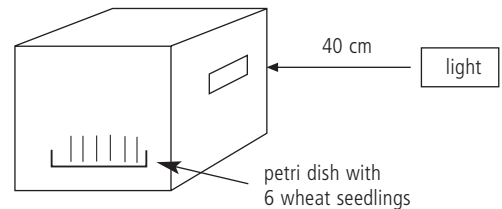


Figure 3: Box set up

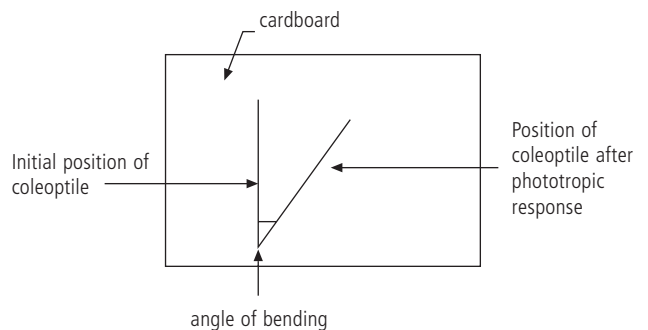


Figure 4: Measuring the angle of bending

Results

Table 2

Colour filters	Wavelength of light (nanometres)	Angle of bending
		24 hours
Clear	400–700 nm	
Red	650–700 nm	
Yellow	560–600 nm	
Green	500–560 nm	
Blue	430–500 nm	

Analysis of results

- 1 Why was clear filter used?
- 2 Under which coloured filters did the seedlings bend the most? Explain why.
- 3 Did your findings support your hypothesis?

Part C – Geotropism in lima bean seeds

Aim To observe a geotrophic response in lima bean seeds.

Hypothesis

Write a hypothesis predicting how roots will grow from seeds placed in different positions.

- Equipment**
- 8 lima bean seeds
 - petri dish
 - vermiculite
 - filter paper
 - scissors
 - tap water
 - eye dropper
 - masking tape
 - water-proof pen
 - saucer
 - 2% bleach solution
 - small bowl
 - modelling clay
 - a long nail
 - tongs
 - Bunsen burner
 - hot mat
 - hot gloves

Method

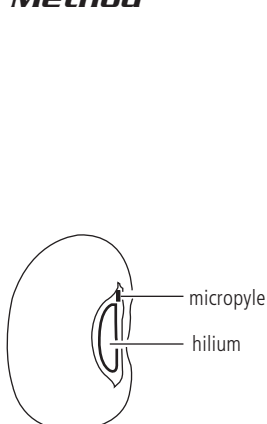


Figure 5: Parts of the bean seed

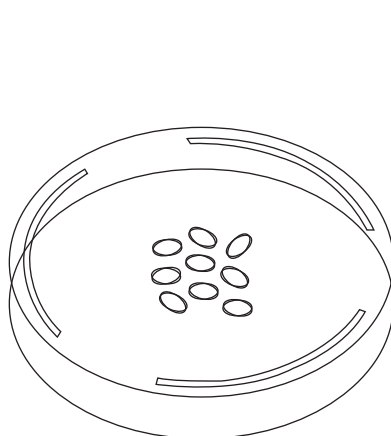


Figure 6: Air holes in the petri dish

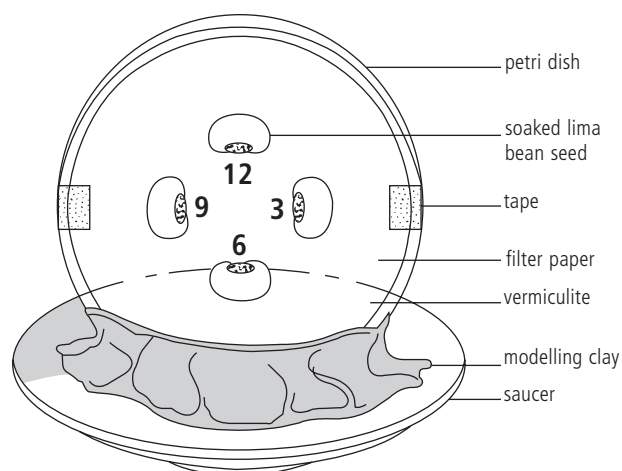


Figure 7: Experimental set up

The **hilum** is the point at which the seed was attached to the pod.
The **micropyle** is the pore through which water enters the seed when it is soaking.

- 1 Place the bean seeds in the small bowl and soak them in water overnight.
- 2 The next day, soak the bean seeds in 2% bleach solution for 30 minutes to inhibit the growth of mould on the seeds.
- 3 Make air holes in the bottom of the petri dish: Work in a well-ventilated area. Place the dish upside down. Wearing hot gloves, heat a nail by holding it with tongs over the blue flame of the Bunsen burner. When the nail is hot, force it through the bottom of the dish. Make 9 holes. (See figure 6.)
Note: Your teacher may provide petri dishes perforated with holes.
- 4 Place a uniform layer of vermiculite in the bottom of the petri dish, over the air holes. Cut the filter paper to the size of the petri dish, and use it to cover the vermiculite.
- 5 Identify the hilum and the micropyle on the bean seeds. (See figure 5.)
- 6 Choose four well-formed seeds. Place them in the dish on top of the vermiculite in the positions shown in Figure 8: at 12 o'clock, 9 o'clock, 6 o'clock and 3 o'clock positions.
- 7 Add water to the seeds and place the cover on the petri dish. Make sure that the seeds are held in place by the cover when the dish is placed on its end. If they are not held in place, add more vermiculite. Tape the cover onto the dish.
- 8 Draw arrows on the dish with the marker to indicate the orientation of the dish to be maintained throughout the experiment. (See figure 7.)
- 9 Place a small amount of modelling clay in a glass saucer. Set the petri dish on its end in the modelling clay, as shown in figure 7. Use a waterproof pen to write the name of your group on the side of the petri dish.
- 10 Place the dish in a dark cupboard. **Note:** The dishes are kept in a dark cupboard so that the only variable being tested is the seeds' response to gravity. Their response to light is not being tested.
- 11 Check the seeds each day for 5 days. Record your observations in Table 3 and draw diagrams to show the orientation of roots from each seed.
- 12 After 5 days, dispose of the seeds in the rubbish bin. Clean and pack up the equipment.

Results

Table 3 Direction of root growth

Day of observation	Seed position			
	12	9	6	3
1				
2				
3				
4				
5				

- 1 Draw diagrams to show how the roots grew.

Analysis of results

- 1 In what direction did the roots grow for each of the bean seeds?
- 2 What type of response was shown by the roots of the four bean seeds?
- 3 How is a plant helped by the response of a root to gravity?

Conclusion

- 1 Define phototropism and geotropism.
- 2 Explain the mechanism of phototropism and geotropism.
- 3 How is phototropism affected by different wavelengths of light?
- 4 How does the position of the seed affect the growth of roots?

INVESTIGATING REFLEX ACTIONS



Background

Timing: 2 × 45 minute lessons

The **brain** controls **voluntary actions** which are under our conscious control. The **brain stem** controls **involuntary reflexes** such as breathing, heart beat and blood pressure. The **midbrain** and the **spinal cord** function as the integration centres for many reflexes. **Reflexes** are rapid, involuntary responses to stimuli which are mediated over simple nerve pathways called **reflex arcs**. Involuntary reflexes are very fast. The fastest impulses can travel at 520 kilometres per hour. Sneezing, coughing, and blinking are simple reflexes.

Reflexes serve as primitive responses that protect the human body from danger and help it to adjust to the surroundings. For example, humans cough when an irritant enters the windpipe and needs to be expelled through the mouth. We sneeze when we need to clear irritants and allergens from our nasal air passages. We blink when danger threatens the sensitive tissues of the eye and when we need to moisten and clean the cornea. (This reflex occurs 900 times an hour!) We yawn when nerves in the brain stem find that there is too much carbon dioxide in the blood. A yawn makes the muscles in the mouth and throat contract and forces the mouth wide open, allowing us to expel carbon dioxide and take in a large amount of oxygen-rich air. When we put a finger on a hot stove, we immediately remove the finger before the message, 'My finger is on a hot burning stove,' arrives in the brain.

Reflex testing is an important diagnostic tool for assessing the condition of the nervous system. Distorted, exaggerated or absent reflex responses may indicate degeneration or pathology of portions of the nervous system, often before other signs are apparent. If the spinal cord is damaged, then reflex tests can help determine the area of injury.

SAFETY RULES

- ⚠ Be careful when you are doing reflex action experiments. You must not throw the cotton ball directly at the subject's eyes, shine the torch at the subject's eyes for too long, hit the subject's knee too hard.

Part A – Measuring the speed of a nerve impulse and the reaction time

Aim To investigate how the speed of a nerve impulse varies according to the visual and auditory inputs given to the subject to catch a falling ruler.

Equipment • metre ruler • rubber hammer • table and chair

This reaction time experiment requires visual information (the movement of the ruler) to travel to your brain. Then your brain sends a motor command (catch that falling ruler) to the muscles of your arm and hand, which then catch the ruler. The nerve impulses in the neurons that control these reflexes travel an express route.

Method

Eyes open

- 1 Work in pairs. One student is the subject, the other student is the experimenter.
- 2 The subject sits with their forearm resting on a table surface, so that the hand extends over the edge of the table. The experimenter stands, holding the top of the metre ruler with the zero end between the thumb and fingers.
- 3 The lower (100 cm) end of the ruler should be between the subject's forefinger and thumb of the **dominant hand**, ready to catch it.
- 4 The subject keeps their eyes open and catches the ruler when the experimenter drops it.
- 6 The experimenter records the distance the metre ruler falls in Table 1.
- 7 Repeat the experiment three more times (four times in total), record each distance and calculate the average distance.
- 8 Have the subject switch to the **non-dominant hand** and repeat the experiment four times. Record the results in Table 1. Note if there is any difference in reaction time between the dominant and non-dominant hand.

Eyes closed

- 9 Have the subject close their eyes. The experimenter taps the subject on the head at the same moment the ruler is dropped. The tap on the head is the subject's signal to catch the ruler with their dominant hand. Repeat four times in total, recording the average distance in Table 1.
- 10 Have the subject close their eyes. The experimenter taps the subject on the non-catching hand at the same moment the ruler is dropped. The tap on the hand is the subject's signal to catch the ruler with their dominant hand. Repeat four times in total, recording the average distance in Table 1.
- 11 Have the subject close their eyes. The experimenter taps the subject on the foot at the same moment the ruler is dropped. The tap on the foot is the subject's signal to catch the ruler with their dominant hand. Repeat four times in total, recording the average distance in Table 1.
- 12 Use the table below to calculate the reaction time in seconds.

Distance of catch (cm)	5	10	15	20	25	30	43	61	79	99
Reaction time (seconds)	0.10	0.14	0.17	0.20	0.23	0.25	0.30	0.35	0.40	0.45

Note: if you catch the ruler at the 20 cm mark, then your reaction time is equal to 0.20 seconds (200 ms). Remember that there are 1000 milliseconds (ms) in 1 second.

Results

Table 1

Trials		Distance of drop (cms)					
		1	2	3	4	Average	Reaction times (ms)
Eyes open	Dominant hand						
	Non-dominant hand						
Eyes closed	Tap on head						
	Tap on hand						
	Tap on foot						

Analysis of results

- 1 Did the reaction time of the subject decrease or increase over 4 trials?
- 2 Did you notice any difference between the dominant and non-dominant reaction time?
- 3 How did the reaction speeds compare for the three experiments where the subject's eyes were closed?
- 4 Is this reaction of catching a ruler voluntary or involuntary? Explain.

Part B – Investigating reflex actions

Aim To investigate the blink reflex, pupillary response to light and knee-jerk reflex.

Equipment

- a clear plastic sheet
- plastic ruler
- cotton ball
- torch

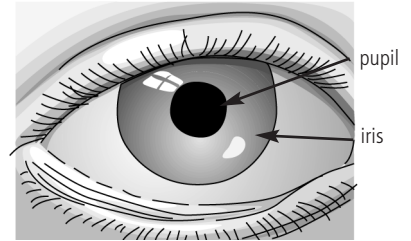
Method**Blink reflex**

The pupillary reflexes are autonomic reflexes that can be used to assess multiple conditions, such as deterioration of vital brain tissue and severe head trauma.

- 1 Have the subject use both hands to hold a thin sheet of clear plastic in front of their eyes so that their nose is touching it.
- 2 The experimenter must toss a cotton ball at the plastic sheet, aiming gently at the subject's eyes. Record their reaction in Table 2.

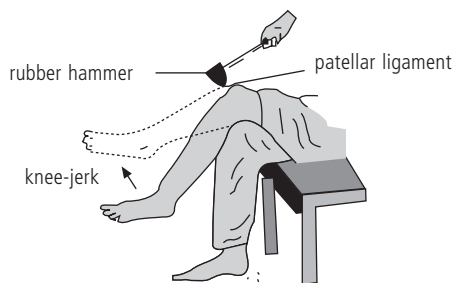
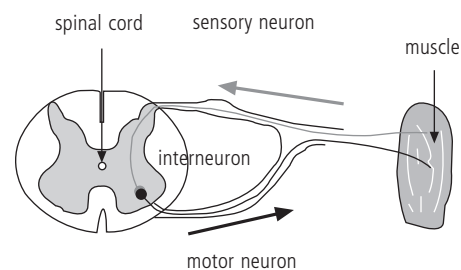
Reaction to light: photopupillary reflex

- 3 Dim the lights in the room. After a few minutes, look at the subject's eyes and measure the diameter of the pupil (the black centre spot in the middle of the eye).
- 4 Shine a torch directly into the subject's eyes. Measure the size of the pupil again. Record your results in Table 2.

**Figure 2: Contraction of the pupil****Knee-jerk reflex (patellar reflex)**

The knee-jerk reflex is a falling reflex. When we are off balance, this reflex 'kicks' us into the air to help us recover our footing. The tap below the knee causes the thigh muscle to stretch. Information is then sent to the **spinal cord** via the **sensory nerve**, which is connected to the **motor nerve** by the **interneuron**. The knee-jerk reflex takes about 50 milliseconds from the tap below the knee to the start of the leg kick. That is fast!

- 5 The subject sits on a chair with their legs hanging freely, or one knee crossed over the other.
- 6 The experimenter uses a rubber hammer or the side of their hand to hit the subject's leg just below the **kneecap** (patella). Observe and record what happens to the subject's leg. Record the time in seconds taken for the knee jerk to occur in Table 2.

**Figure 3: Knee-jerk test****Figure 4: Reflex arc****Results****Table 2**

Reflex measured	Observation	Interpretation
Blink reflex		
Pupillary action	Size of the pupil before shining: ____ mm Size of the pupil after shining: ____ mm	
Knee reflex	Time taken for the knee to jerk: ____s	

Analysis of results

- 1 What did the subject do when the cotton ball was thrown at their eyes? Why did this reflex occur?
- 2 Why does the size of the pupil change in the dark and the light?
- 3 What happened when the subject's knee was tapped?
- 4 How does the brain receive the message from the sense organs (eyes, ears and skin)?
- 5 How does the brain send the message back to the muscles to grasp the ruler?
- 6 Where along a reflex arc would you find interneurons? What are the functions of the interneurons?

Conclusion

- 1 Explain why the subject reacted faster when their eyes were open than when their eyes were closed.
- 2 What is a reflex action? Which part of the central nervous system controls reflex actions?
- 3 What is the purpose of a reflex action?
- 4 Why is reflex testing an important diagnostic tool in physical examinations of patients by physicians?

Background **Timing:** 2 × 45 minute lessons

Receptors monitor changes in the external and internal environment and send nerve impulses via **sensory nerves** to the **sensory cortex** of the brain. The brain interprets the stimulus and sends nerve impulse by **motor nerves** to the muscles or glands (**effectors**) to bring about a response.

The **skin** is divided into the outer **epidermis** and the **inner dermis**. The skin has touch, pain, pressure, vibration and temperature receptors. Every square centimetre of your skin contains around 200 pain receptors, but only 15 receptors for pressure, 6 for cold and 1 for warmth. These receptors are not evenly distributed. They are close together on the finger tips and face, and far apart on the back, legs, arms and stomach.

The cold and warm **temperature receptors** adapt and become less sensitive if a stimulus is continued. For example, if one hand is placed in warm water and the other is placed in cold water, the temperature receptors will become less sensitive. If both hands are placed in lukewarm water immediately afterwards, the hand originally in warm water will feel cold, and the hand originally in cold water will feel warm. This adaptation allows us to tune out background noise, to ignore the touch sensation from our clothing and to lose awareness of the temperature of the immediate environment.

Neurologists use two-point discrimination tests to check for nerve damage. They may do this if they suspect a problem with the entry of sensory information to the skin, the pathways to the brain, or the interpretation of sensory information.

- Aim**
- To investigate temperature and touch receptors in the skin.
 - To predict and then measure a person’s ability to discriminate between one point and two points touching various places on the skin.

- Equipment**
- two identical 20-cent coins
 - 3 × 200 mL beakers
 - ice
 - Bunsen burner
 - tripod
 - gauze mat
 - heat mat
 - matches
 - thermometer
 - 8 corks
 - tooth picks
 - metric ruler
 - water
 - paper towel
 - blindfold
 - graph paper

SAFETY RULES

- Take precautions to prevent the spreading of germs: wash your hands, dispose of chemicals properly and use the equipment properly.
- Make sure that the water used is at the specified temperature. Do not use boiling water and do not leave your finger in the cold water for too long.
- Be careful when testing two-point discrimination of skin areas. Do not apply too much pressure on the skin. Use fresh toothpicks and apply the points of the toothpicks gently.
- To avoid the spread of germs, do not share the blindfolds.

Sense receptors in the skin

Sensory nerves carry impulses from the skin receptors to the sensory cortex of the brain. The motor nerves carry the message back to the effectors, such as the muscles and glands

Types of receptors in the skin

Name	Location	Function
Hair follicle endings	Hairy skin areas	Respond to hair displacement
Ruffini endings	Located in middle dermis below the cold receptors of hairy and hairless skin	Respond to heat and pressure on skin
Krause corpuscle	Close to epidermis	Responds to cold temperatures
Pacinian corpuscle	Deep layers of dermis in hairy and hairless skin	Responds to high frequency vibrations in 150–300 Hz range
Meissner corpuscle	Located in the hairless area of the upper dermis	Responds to low frequency vibrations in 20–40 Hz range

Name	Location	Function
Free nerve endings	Throughout the skin	Different types of free nerve endings respond to mechanical, thermal and noxious stimulation.
Merkel's nerve complex	Epidermis of skin	Responds to steady pressure on the skin
Heat and cold receptors	Located in upper dermis of both hairy and hairless areas	Responds to changes in temperature
Pain receptors:	Located around the base of hairs near the epidermis	Respond to mechanical trauma, very high and very low temperatures

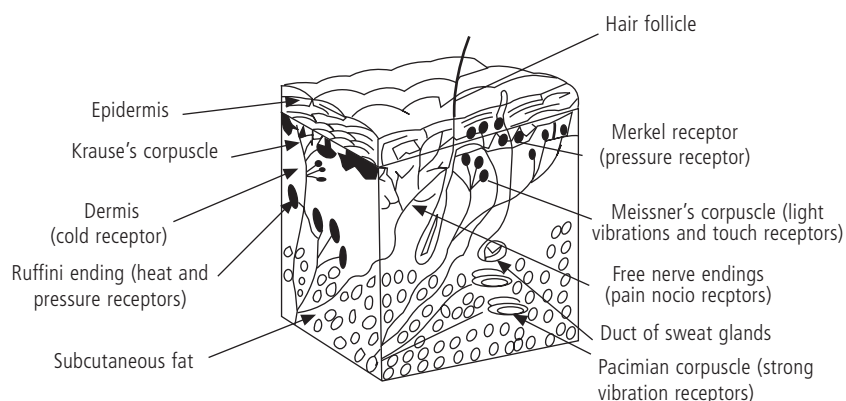


Figure 1: Location of skin receptors

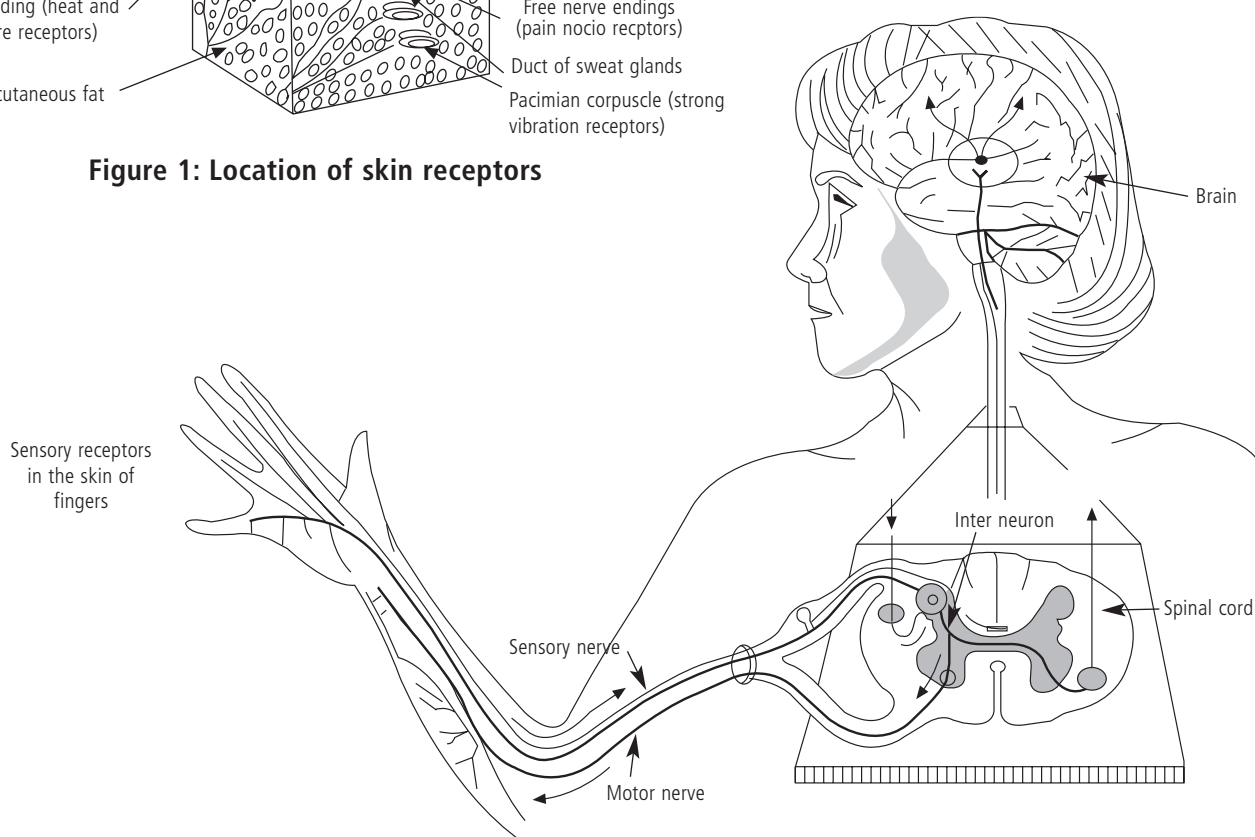


Figure 2: Pathway showing how sensations from the skin are perceived

Part A – Investigation of heat and cold receptors

Method

- 1 Half fill one beaker with ice water (0°C), the second beaker with water at room temperature (25°C) and the third beaker with warm water (50°C). Arrange them in a row from coolest to warmest.
- 2 Have the subject put their right forefinger into the cold water and their left forefinger into the warm water. Leave the fingers immersed for one minute.
- 3 After one minute, have the subject quickly immerse both forefingers in the middle (room temperature) beaker. Ask the subject what sensation they feel in each finger. Record the results in Table 1.

Results

Table 1 Sensations felt by forefingers in water

Finger used	Sensation felt: hot, warm or cold		
	Cold water	Water at room temperature	Warm water
Right forefinger			
Left forefinger			

Analysis of results

- 1 Describe how the subject felt when their right finger was immersed in cold water and then in the water at room temperature.
- 2 Describe how the subject felt when their left finger was immersed in warm water and then in water at room temperature.
- 3 Explain the reason for this discrepancy by the receptors in the fingers in feeling the correct temperature.

Part B – Investigation of touch receptors

Method

Perception of objects

- 1 Work in groups of three. One student is the **subject**, the second student is the **experimenter** and the third student is the **recorder**.
- 2 Have the subject place their head on the bench, face upwards and close their eyes.
- 3 The experimenter places two identical 20-cent coins about 4 cm apart on the subject's forehead. Ask the subject whether one coin is heavier than the other, or both feel the same weight.
- 4 The recorder notes their response under Results.

Two-point discrimination

- 5 Have the subject close their eyes or wear a blindfold. (The subject is not allowed to watch the procedure.)
- 6 The experimenter should place the blunt ends of two toothpicks in a cork, 50 mm apart and inserted to equal depths. Holding the cork, place the toothpicks on the subject's **forearm**. Make sure that the two points touch their skin at the same time. Ask the subject how many points they feel.
- 7 If the subject feels two points, move the toothpicks closer together so that they are 45 mm apart. Repeat the test.
- 8 Continue the procedure until you find the smallest distance between the toothpicks at which the subject feels two points instead of one. When the subject first reports one point, move the toothpicks apart by two millimetres at a time. Continue until the subject feels two points again.
- 9 When you have found the smallest distance, have the recorder measure the distance in millimetres between the two toothpicks. Record this distance and your observations about the sensitivity of the skin area in Table 2.
- 10 Repeat the procedure for the other skin areas shown in Table 2.
- 11 Collate the class results.

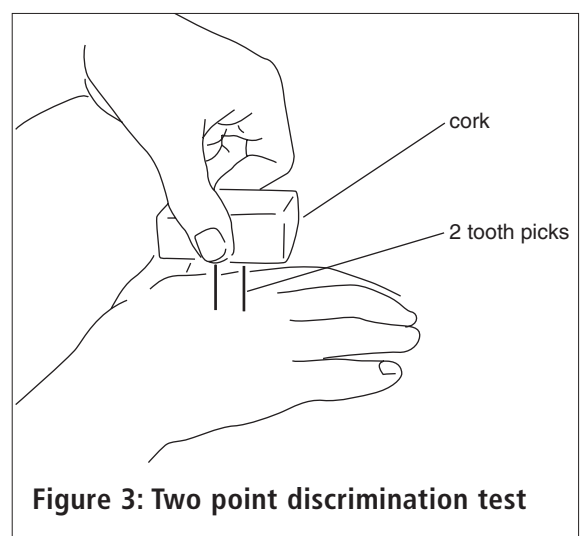


Figure 3: Two point discrimination test

Results**Perception of objects**

- 1 What was the subject's perception about the weight of the coins?

Table 2 Two point discrimination

Areas of skin tested	Expected two-point distance	Actual two-point distance	Sensitivity
Forearm	35 mm		
Finger tip	2–3 mm		
Upper lip	5 mm		
Cheek	6 mm		
Nose	7 mm		
Palm	10 mm		
Forehead	15 mm		
Upper arm	39 mm		

- 2 Draw a histogram to show your results. List the areas of skin that you tested on the x-axis and the two-point distance on the y-axis.

Analysis of results

- 1 How do your results compare with the results of other groups? Can you explain the differences you see among the observations?
- 2 Which parts of the body are best at recognising two points even when the points are very close together?
- 3 Which skin areas do you think have more receptors: areas that have small two-point distances or areas that have large two-point distances? Why?
- 4 Which brain area do you think is larger: the area receiving information from skin that has many receptors or the area receiving information from skin that has few receptors?
- 5 How does information from sensory receptors in the skin get to the brain?
- 6 List ways you could improve this experiment.

Conclusion

- 1 Which sensory receptors detected the following sensations?
 - the weight of the coins
 - the temperature of water
 - two-point discrimination
- 2 Which part of the body is the most sensitive to two-point discrimination?
- 3 Describe the location in the skin of the touch/pressure receptors, pain receptors, and heat and cold receptors. (Refer to the skin diagram in Figure 1.)

ORIENTATION MOVEMENTS EXHIBITED BY PLANARIA



Timing: 2 × 45 minute lessons

Background

Planaria (*Dugesia tigrina*) are common freshwater flatworms. They are usually found under rocks in streams and ponds. Planaria are members of the phylum **Platyhelminthes** (flatworms) in the class **Turbellaria**. They glide about using **cilia**, which are located on their lower surface. The **two 'eye' spots** near the front end allow the planarian to sense light levels in its environment. Two types of movements are exhibited by planaria.

Kinesis is a non-directed movement in response to environmental factors such as light, temperature, moisture or chemical cues. When an organism is in an unfavorable location, it may increase its speed, angle of movement or amount of turning. If effective, the actions will remove the organism from the adverse location. Kinesis results in random movement into better locations.

Taxis involves directed movements; that is, animals detect the source or differences in intensity of a factor (such as temperature), then they move accordingly. A **positive taxis** is a movement towards a stimulus; a **negative taxis** is a movement away from a stimulus. The response to light is **negative phototaxis**, response to gravity is **positive geotaxis** and the response to chemicals is **negative chemotaxis**. When the organism is touched, it contracts and exhibits **thigmotaxis**.

Aim

To observe phototaxis, thigmotaxis, geotaxis and chemotaxis in planaria when they are exposed to light, touch, gravity and chemicals.

Equipment

- test tube
- test tube rack
- 2% vinegar solution
- 2% salt solution
- pond water
- planarian culture (20 planaria per group)
- pipette
- waterproof pen
- 2 petri dishes
- 2 lamps
- cardboard measuring 2 cm × 3 cm
- paper towel

SAFETY RULES

- ⚠ Wear a laboratory coat and gloves while doing this experiment.
- ⚠ You may find it difficult to touch a planarian with the pipette. They do tend to attach themselves to the glass tube. Handle them carefully.
- ⚠ Do not add too much salt or vinegar as this could kill the planarian.
- ⚠ After the experiment, return the planaria to a pond or stream.

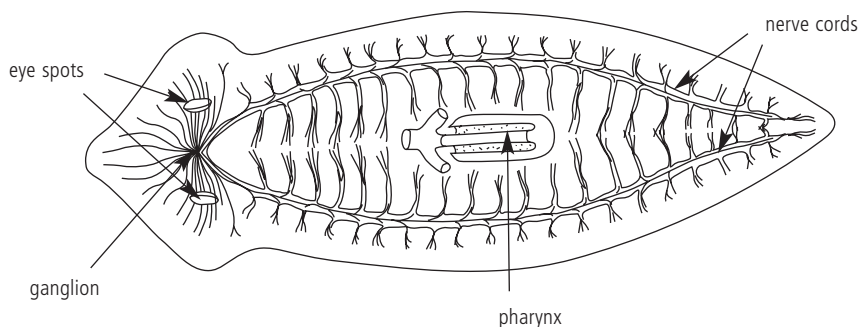


Figure 1: Structure of a planarian

Part A – Reaction to light and touch: phototaxis and thigmotaxis

Hypothesis

Write a hypothesis to predict how planaria will react to light and touch.

Method

- 1 Label two petri dishes A and B. Pour pond water into both petri dishes and place 10 planaria in each dish. **Note:** Keep the room lighting low during this experiment. Stray light can cause unexpected results.
- 2 Place a lamp above each petri dish. Make sure that light is not close enough to heat the water.
- 3 Use the cardboard to completely cover petri dish B. Leave it in place for 3 minutes.
- 4 Note the position of the planaria in each dish.
- 5 After the planaria begin to swim, touch one gently with a pipette. Observe the response. If the planarian contracts, it is showing thigmotaxis (response to touch).
- 6 Slide the cardboard across dish B so that half of the dish is exposed to the light. (See figure 2.)
- 7 Draw diagrams to record the position of planaria in dish A and dish B.
- 8 Count the number of planaria in the lighted area in dish B.
- 9 Note the speed and direction of the planarian movement in each dish.
- 10 Continue your observation until all the planaria in dish B are under the cardboard.
- 11 Slide the cardboard cover across petri dish B so that the lighted side is covered and the dark side is exposed to the light source. Continue observation until all the planaria in dish B are under the cardboard.
- 12 Repeat the procedure until both area 1 and area 2 of dish B have been exposed to light twice. You should have 4 sets of readings of planarian movement.

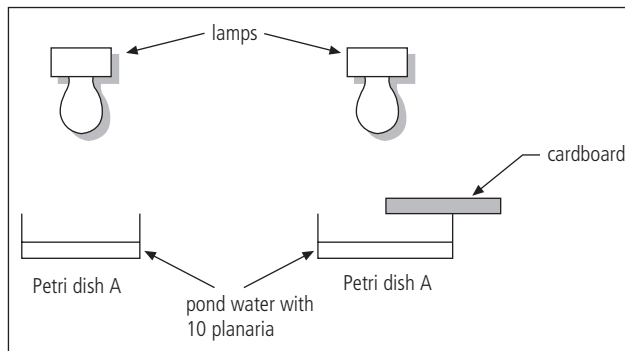


Figure 2: Part A experimental set up (phototaxis)

Results

Table 1 Reaction to light

Observation	Dish A	Dish B	
	Lighted constantly	Area 1 lighted	Area 2 lighted
Number of planaria			
Movement of planaria (speed and direction)			

Reaction to touch: When a planarian is touched with a pipette it _____ and exhibits _____.

Part B – Reaction to gravity: geotaxis

Hypothesis

Write a hypothesis to predict how planaria will react to gravity.

Method

- 1 Fill a test tube with pond water and place it in an upright position in a test tube rack. Draw a line across the test tube, half way between the bottom of the tube and the water level.
- 2 Place one planarian in the test tube.
- 3 Wait until the planarian reaches the centre line you drew. Observe for 3 minutes, noting whether it moves down (towards gravity) or up (away from gravity). Record your results in Table 2.

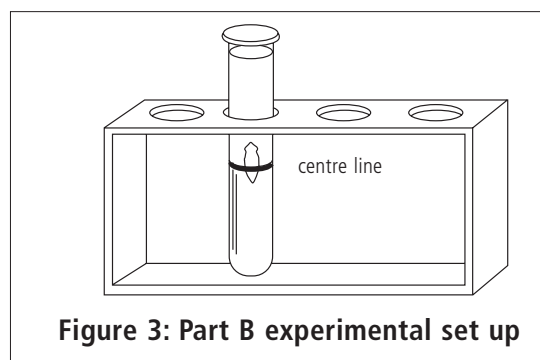


Figure 3: Part B experimental set up

Results

Table 2 Planarian reaction to gravity

Time spent in the bottom part of the test tube (mins)	Time spent at top part of the test tube (mins)	Time spent at the middle of the test tube (mins)	Geo taxis: positive or negative

Part C – Reaction to chemicals: chemotaxis

Hypothesis

Write a hypothesis to predict how planaria will react when exposed to chemicals.

Method

- 1 Use a waterproof pen to draw a line across the middle of the bottom of a petri dish. Fill the dish with pond water and add 5 planaria.
- 2 Wait for the planaria to reach the centre line. Then add one drop of 2% vinegar to the left side of the petri dish. **Note:** Too much vinegar will kill the planaria.
- 3 Observe the planaria for 3 minutes, noting whether they move towards or away from the vinegar. Record your results in Table 3.
- 4 Wait for the planarian to reach the centre line again. Then add one drop of 2% salt solution to the right side of the dish.
- 5 Observe the planaria for 3 minutes, noting whether they move towards or away from the salt solution. Record your results in Table 3. Note: Too much salt solution will kill the planaria.

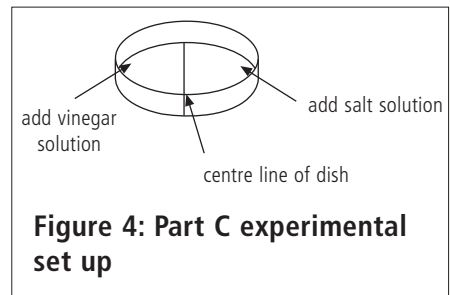


Table 3 Reaction to chemicals

Time spent at the centre line (mins)	Time spent near vinegar (mins)	Time spent away from vinegar (mins)	Time spent near salt solution (mins)	Time spent away from salt solution (mins)	Chemotaxis: positive or negative

Analysis of results

Collate the class results for parts A, B and C. Then answer the questions.

- 1 Which dish was used as the control in part A?
- 2 Why was dish B partly covered with cardboard?
- 3 What happened when a planarian was touched with a pipette?
- 4 In which direction did the planaria move when exposed to light? Gravity? Vinegar and salt?
- 5 To which environmental stimuli did the planaria react positively? Negatively?
- 6 What is the advantage of collating class results?
- 7 What problems did you encounter in this experiment?
- 8 How could you modify the experiment to minimise experimental errors?

Conclusion

What types of orientation movements were exhibited by the planaria when they were exposed to light, touch, gravity and chemicals?

SOCIAL AND AGGRESSIVE BEHAVIOUR IN FISH



Timing: 2 × 45 minute lessons

Background

Zebra fish (*Brachydanio rerio*) are small freshwater fish. They swim together in a group or school. A **fish school** is defined as a group in which the animals stay together because they are showing positive social responses to each other; not because they are responding similarly to a common external factor such as light and food. The major benefits of social behaviour are the ability of the group to defend against competitors and predators; the greater chance of a group member noticing a predator and warning the others; the reduced chance of an individual being captured by a predator; the greater chance of finding food; greater efficiency in capturing prey when foraging.

However, there are disadvantages of social living, in the greater chance of being detected by a predator, greater food competition within the group and a greater chance of diseases being transmitted among the group.

Goldfish (*Carassius auratus*) and **catfish** (*Synodontis granulatus*) also exhibit schooling behaviour. Male **Siamese fighting fish** (*Betta splendens*) are not very social. They are very aggressive towards other males and can kill each other, especially during mating season. The winner mates with the females.

Aim

- To investigate the social and schooling behaviour of zebra fish and goldfish towards their own species and different species.
- To investigate the social and aggressive behaviour of male catfish and male Siamese fighting fish towards their own species and different species.

Equipment

- 2 male catfish
- 2 male Siamese fighting fish
- 10 male zebra fish
- 10 small male goldfish
- a net
- a large glass tank
- three opaque plastic dividers
- four 1-litre glass jars (same depth as the tank)

SAFETY RULES

- Wear lab coats to hide brightly coloured clothes because these colours may excite the fish.
- While handling live animals, be careful not to stress them by being playful or aggressive.
- Do not tap on the glass tank or talk while observing their behaviour.
- Avoid abrupt movements when you are near the fish and speak quietly.
- If the fish are over-stimulated, they will become used to the stimulus and may not respond during the later parts of the experiment.

Part A – The tendency of fish to school with their own species

Method

- 1 Draw two vertical lines on the aquarium front wall to divide it into three equal areas.
- 2 Using jars of the same size, place 6 zebra fish in one jar and 6 goldfish in a second jar.
- 3 Place the jars at opposite ends of the aquarium. Fill the aquarium with water.
- 4 Place one zebra fish (test fish 1) in the middle of the aquarium. Observe its movements for 4 minutes, noting the time (in minutes) the fish spends in each of the three areas and the number of times it moves from one area to another. Record your observations in Table 1.
- 5 Reverse the positions of the jars and repeat the test.
- 6 Repeat steps 4 and 5 with a second zebra fish.
- 7 Replace the zebra fish with a goldfish and repeat steps 4, 5 and 6.

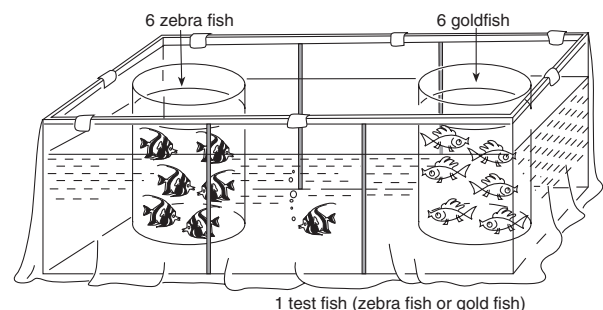


Figure 1: Part A experimental set up

Results

Table 1

	Zebra fish			Gold fish		
	1	2	Average	1	2	Average
Time spent with same species (mins)						
Time spent in the middle (mins)						
Time spent with different species (mins)						

Part B – Shoal number preference

Method

- 1 Put two zebra fish in one jar. Put six zebra fish in the other jar. Place the jars at opposite ends of the aquarium.
- 2 Put a test zebra fish in the centre of the aquarium. Observe its movements for 4 minutes, noting the time (in minutes) the fish spends in each of the three areas and the number of times it moves from one area to another. Record your observations in Table 2.
- 3 Reverse the positions of the jars and repeat the test.
- 4 Repeat steps 2 and 3 with a second zebra fish.

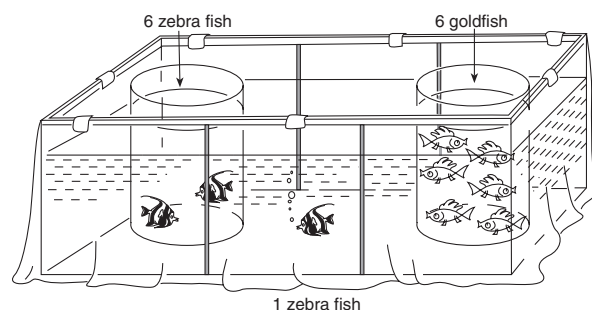


Figure 2: Part B experimental set up

Results

Table 2

	Zebra fish		
	1	2	Average
Time spent in area with 2 fish (mins)			
Time spent in area with 6 fish (mins)			
Time spent in the middle (mins)			

Part C – The social and aggressive behaviour of catfish and Siamese fighting fish

Method

- 1 Fill 4 one-litre jars with warm water (27–28°C, pH = 7).
- 2 Use three opaque plastic dividers to separate the 4 jars so that the fish will be unable to see each other.
- 3 Place one male catfish in each of two jars (jar 1 and jar 2). Place one male Siamese fighting fish in the third and fourth jars (jar 3 and jar 4). Allow them to acclimatise to the water for **two days**. Maintain the water temperature at 27–28°C and pH of 7.

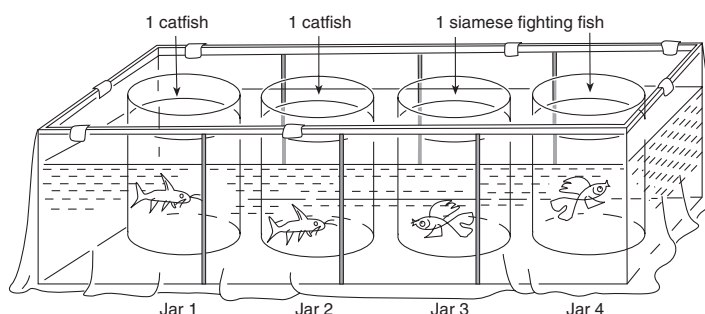


Figure 3: Part C experimental set up

Social behaviour of catfish

- 4 Remove the opaque screen between jar 1 and jar 2 (the two jars containing the catfish).
- 5 Record the behaviour of each fish when they see each other. Note the position of the **tail, fins, mouth** and **gill covers**.

- 6 Replace the opaque divider between the two fish. Record your results in Table 3.

Aggressive behaviour of Siamese fighting fish

- 7 Remove the opaque divider between jar 3 and jar 4 (the two jars containing the Siamese fighting fish).
- 8 Record the behaviour of each fish when they see each other. Note the position of the **tail, fins, mouth** and **gill covers**.
- 9 Replace the opaque divider. Record your results in Table 3.

Catfish and Siamese fighting fish social behaviour

- 10 Remove the opaque screen between jar 2 and jar 3 (one jar containing a catfish, one jar containing a Siamese fighting fish).
- 11 Record the behaviour of each fish when they see each other. Note the position of the **tail, fins, mouth, gill covers** and the **colour change**.
- 12 Replace the opaque divider. Record your results in Table 3.

Results

Table 3 Social and aggressive behaviour

Behaviour pattern	Cat fish with catfish	Siamese fighting fish with Siamese fighting fish	Cat fish with Siamese fighting fish
Position of fins			
Position of tail			
Movement of mouth			
Movement of gill covers			
Colour change			

Analysis of results

- 1 Did the test zebra fish spend more time near the jar of goldfish or near the jar of zebra fish?
- 2 Did the test zebra fish show a preference for a small shoal or a larger shoal?
- 3 Did the test zebra fish show a preference towards its own species or another species?
- 4 Which fish species exhibited more social behaviour towards its own species? Aggressive behaviour towards its own species?
- 5 Why are male Siamese fighting fish and not females used in part C of this experiment?
- 6 How do male Siamese fighting fish recognise each other?

Conclusion

- 1 Summarise the social and aggressive behaviour of zebra fish, goldfish, catfish and Siamese fighting fish with both their own species and with other species.
- 2 Why do fish prefer to live in groups? What are the advantages and disadvantages of living in groups?
- 3 Which fish are more aggressive towards each other? What is the advantage of being aggressive?

Background

Timing: 2 × 45 minute lessons

Somatic cells divide by **mitosis** to replace old cells, to repair damaged cells and for growth. During **mitosis**, the somatic cell divides and produces two identical daughter cells. The parent cells are diploid and the two daughter cells are also diploid. Cell division has two parts, **Cytokinesis** and **Karyokinesis**. Cytokinesis is the division of the cytoplasm of a cell. **Karyokinesis** is the division of the **DNA** in the nucleus of a cell. DNA (DeoxyriboNucleic Acid) is the material inside the nucleus of a cell that carries genetic information.

The five stages of karyokinesis are: interphase, metaphase, metaphase, anaphase and telophase. During **interphase**, the chromosomes are coiled like a ball of wool and are called **chromatin**. The nucleolus, nuclear membrane and 2 centrioles are visible. **Centrioles** consist of a pair of short rods lying just outside the nucleus in animal cells and primitive plants cells. During **prophase**, the nucleolus and nuclear membrane disappear. Each centriole moves to opposite sides of the cell. The chromosomes untangle, shorten and start to replicate, forming two sister chromatids joined at a point called the **centromere**. During **metaphase**, all the chromosomes line up in the middle of the cell, one below the other. **Nuclear spindle** forms from the centrioles and each chromosome is attached to the spindle at the centromere. During **anaphase**, the spindle fibres contract and pull the chromatids of the chromosomes apart. Each chromatid from the chromosome moves to opposite poles of the cell. The cell membrane begins to constrict and divide the cell into two daughter cells. During the final stage, **telophase**, the nucleolus and nuclear membranes appear. Two daughter cells are formed, each with the same number of chromosomes as the parent cell.

During **cytokinesis** in animal cells, the cell pinches in the middle. In plant cells, a **cell plate** forms in the middle. Two diploid daughter cells are formed.

Part A – Examining a prepared slide of an onion root tip

In plants, mitosis takes place in the **apical meristems**. These are normally found at the tips of the stems or roots. The root consists of a root cap, a zone of cell division, a zone of cell elongation, a zone of cell differentiation and a zone of maturation. At the tip of the root there is a thimble-shaped cap called the **root cap**. This protects the root from the soil grains. **Meristem** is the region just behind and partially covered by the root cap. It is in this region that the new cells divide by mitosis. In the **zone of elongation**, the cells become longer. In the **zone of cell differentiation**, cells differentiate in size and shape. In the **region of maturation**, the cells become mature and carry on with various functions.

Aim To observe and identify different stages of mitosis in a slide of an onion root tip.

Equipment

- a slide showing mitosis in an onion root tip
- light microscope
- microscope lamp
- models and charts of mitosis
- reference books
- longitudinal section through an onion root tip

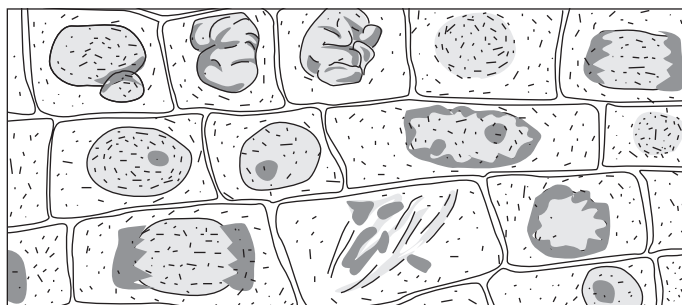


Figure 1: Stages of mitosis in an onion root tip

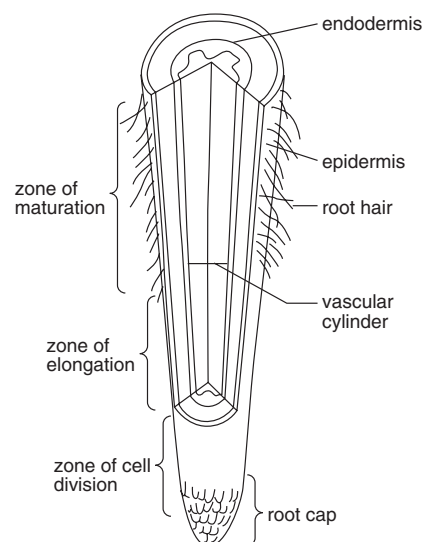


Figure 2: Parts of a root tip

Method

- 1 Use a compound microscope under low-power objective to examine a prepared slide of a longitudinal section through an onion root tip.
- 2 Locate the root cap, zone of cell division, zone of cell elongation and zone of cell differentiation.
- 3 Use high-power objective to observe the zone of cell division where mitosis is taking place. Identify the stages of mitosis: interphase, prophase, metaphase, anaphase, telophase and cytokinesis.
- 4 In Table 1, record the number of cells observed at each stage. Calculate the percentage of cells in each phase.

Results

- 1 Draw and label a diagram of a root tip.

Table 1

Mitotic stages	Number of cells				Percentage of cells Total of cells at this stage ÷ total of cells observed × 100	Time at each stage Percentage × 1440 mins
	Field of view 1	Field of view 2	Field of view 3	Total cells		
Interphase						
Prophase						
Metaphase						
Anaphase						
Telophase						

Graphing

Draw a histogram to show the number of cells in each phase of mitosis. Show the phases of mitosis on the x-axis and the number of cells in each phase on the y-axis.

Analysis of results

- 1 Explain why onion root tips are selected for studying mitosis.
- 2 In which part of the root tip did you observe cells undergoing mitosis?
- 3 What are the functions of the root cap, zone of elongation, zone of cell differentiation and zone of maturation?
- 4 What percentages of cells were seen during each stage of mitosis?
- 5 What difficulties did you encounter when identifying stages of mitosis?

Part B – Modelling stages in Mitosis

Aim To model stages in mitosis using poppit beads and pipe cleaners.

Equipment • 60 white linking beads • butcher's paper • 4 pipe cleaners
• 60 black linking beads • colour pencils or felt pens

Method

- 1 Use 20 white beads to prepare two strands of a long chromosome. Each strand should be made up of 10 beads. Use a pipe cleaner to join the two strands (see figure 3). The pipe cleaner represents the centromere.
- 2 Use 12 white beads to prepare two strands of a short chromosome. Each strand should be made up of 6 beads. Use a pipe cleaner to join the two strands.

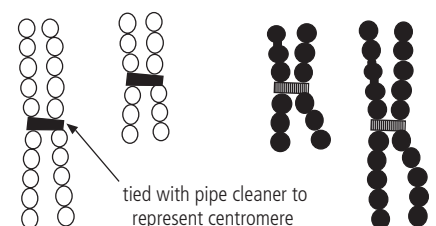
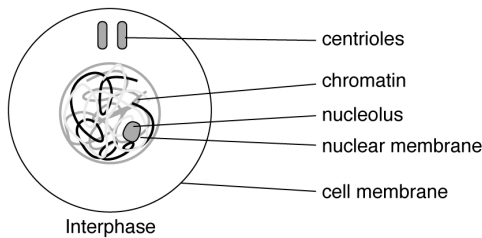
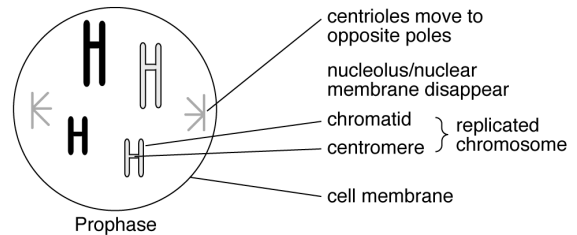


Figure 3: White and black chromosomes

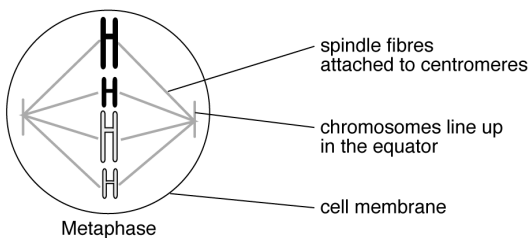
- 3 Repeat steps 1 and 2 using black beads.
- 4 Follow the steps given below to draw your own diagrams on butcher's paper. Use coloured pencils to draw and label the different parts of the cells.



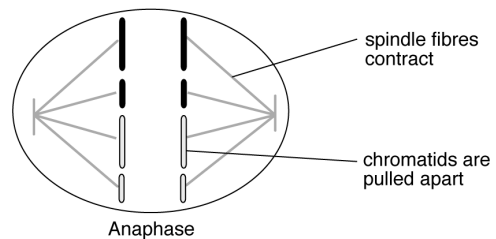
Interphase: Draw a big circle and a smaller circle inside it to represent the nucleus. Draw 2 centrioles above the nucleus. Draw a small nucleolus inside the nucleus. Pile 2 white and 2 black chromosomes in the centre of the nucleus to represent a mass of thread-like **chromatin**. Chromosomes replicate and form chromatids during interphase.



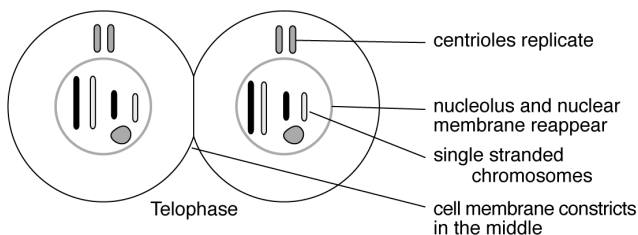
Prophase: Omit drawing the nucleolus and nuclear membrane. Since chromosomes become visible at this stage, lay the 4 chromosomes apart. Draw two centrioles with short spindle fibres at the opposite poles of the cell.



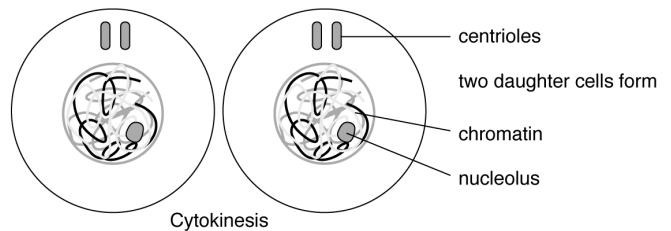
Metaphase: Draw a big circle. Move the chromosomes to the centre of the cell, lining one below the other. Draw spindle fibres coming from the centrioles and join them to the centromeres of each chromosome.



Anaphase: Draw an oval shape. Remove the pipe cleaners. Separate the chromatids and move 4 chromatids to each of the opposite poles of the cell. The chromatids are now called chromosomes.



Telophase: Pile 4 chromosomes at each pole. Draw in nucleoli, nuclear envelope around each new cluster of chromosomes. Draw constriction in the middle of the cell.



Cytokinesis: Draw 2 small circles to represent 2 daughter cells. Draw the nuclear membrane, nucleolus and two centrioles for each daughter cell. Leave the chromosome masses in the centre of the nuclei. In **animal cells** the cell pinches in the middle but in **plant cells** a cell plate forms in the middle. Each daughter cell will have the same number of chromosomes as the parent cell.

Results

Submit your diagrams to your teacher.

Analysis of results

- 1 What is the difference between karyokinesis and cytokinesis?
- 2 Why do somatic cells (body cells) divide by mitosis?
- 3 When do chromosomes align on the equator of the cell?
- 4 How many double-stranded chromosomes are present in the prophase stage?
- 5 How many chromosomes are in each new nucleus of the daughter cells?

Conclusion

Summarise what happens during each stage of mitosis.

Background

Timing: 2 × 45 minute lessons

Meiosis occurs in sex cells to form **gametes** (sperms and ova). Meiosis is divided into two parts: meiosis 1 and meiosis 2. During meiosis 1 and meiosis 2, the cell goes through four stages for each nuclear division: prophase, metaphase, anaphase and telophase. **Meiosis 1** is known as the **reduction division**, because the two daughter cells that are formed at the end of this process receive half the number of chromosomes (**haploid**) of the parent cell (**diploid**). During prophase 1, **homologous chromosomes** (consisting of maternal and paternal chromosomes) pair up (**synapse**). **Crossing over** can occur between two non-sister chromatids and genetic material is exchanged. Crossing over produces genetic variation, the raw material for evolution. **Meiosis 2** is similar to mitosis. At the end of the meiosis 2, four haploid daughter cells are formed which become the gametes. Meiosis provides genetic variation. This gives organisms the ability to adapt to changing environments and also assists the process of evolution.

Aim

To model stages in meiosis using linking beads and pipe cleaners.

Equipment

- 40 white linking beads
- 40 black linking beads
- butcher's papers
- colour pencils or felt pens
- 4 pipe cleaners
- prepared slides of meiosis
- microscope
- models and charts of mitosis
- reference books.

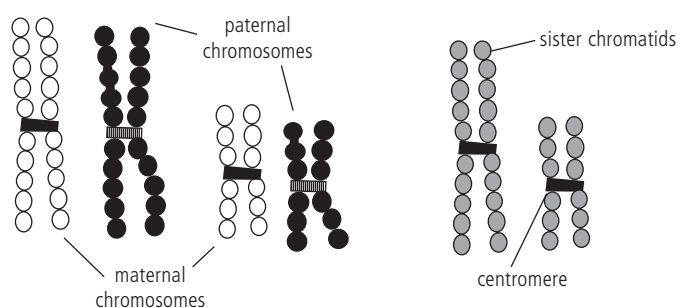


Figure 1: Homologous and non-homologous chromosomes

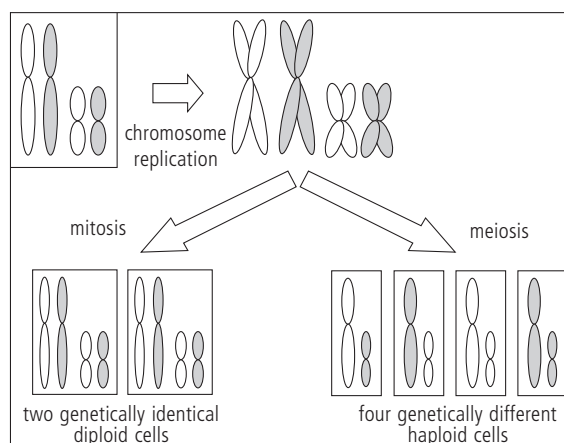


Figure 2: Comparison of mitosis and meiosis

Method

- 1 Use 20 white beads to prepare two strands of a long chromosome. Each strand should be made up of 10 beads. Use a pipe cleaner to join the two strands. The pipe cleaner represents the centromere.
- 2 Use 12 white beads to prepare two strands of a short chromosome. Each strand should be made up of 6 beads. Use a pipe cleaner to join the two strands.
- 3 Repeat steps 1 and 2 using black beads to represent homologous chromosomes.

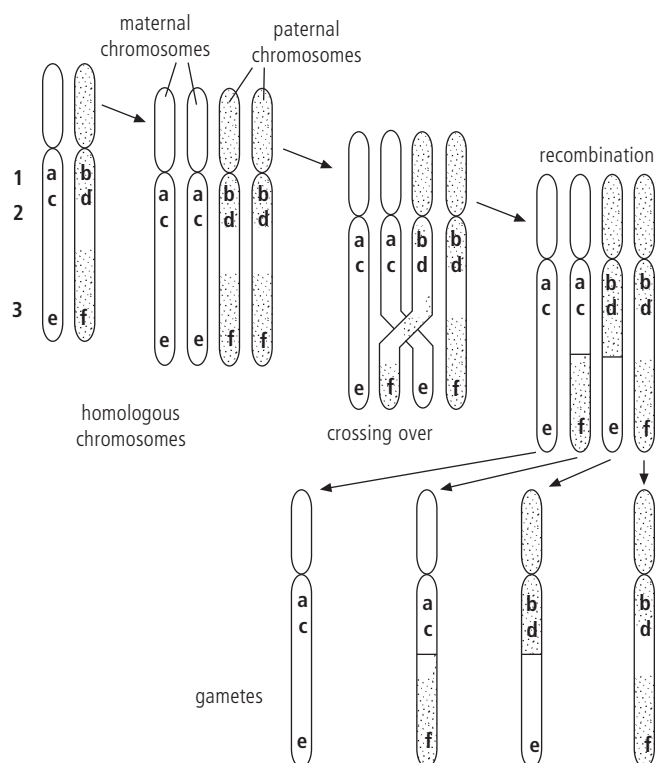
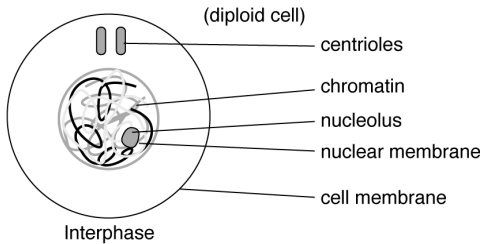


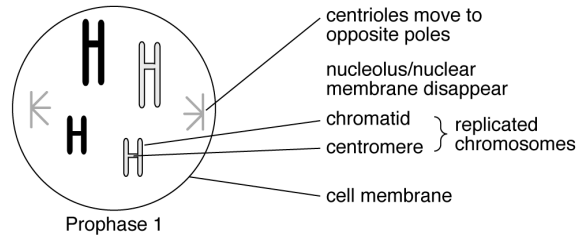
Figure 3: Crossing over and recombination between adjacent homologous chromosomes

- 4 **Meiosis 1:** Divide one piece of butcher's paper into 6 equal sections. Draw 4 large circles in the first 4 sections and label them as interphase 1, prophase 1, metaphase 1 and anaphase 1.
- 5 **Meiosis 2:** Divide the second piece of butcher's paper into 5 sections. Label them prophase 2, metaphase 2, anaphase 2, telophase 2 and cytokinesis 2.
- 6 Following the steps below, complete the diagrams.



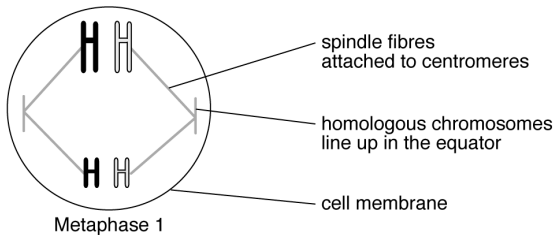
Interphase

Interphase: Pile 4 chromosomes in the centre of the circle to represent a mass of chromatin. Draw 2 centrioles on top of the nucleus. Draw a nucleolus inside the nucleus.



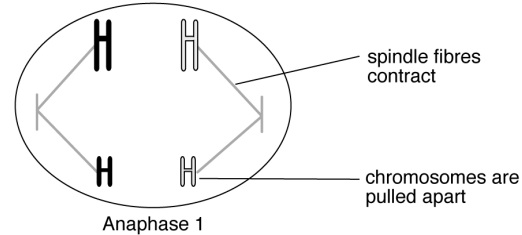
Prophase 1

Prophase 1: Draw the centrioles with short spindle fibres at the opposite ends of the cell. Put two large chromosomes and two small chromosomes next to each other. This represents the pairing of **homologous** chromosomes. Take the inner strand of one white chromosome and **cross over** with the inner strand of its homologous black chromosome. This represents the crossing over of chromatids. Break the chromosome where it crosses over and rejoin the pieces to the opposite chromatids.



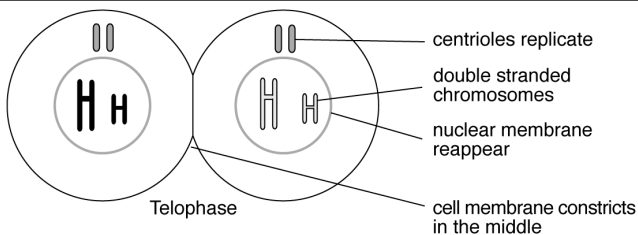
Metaphase 1

Metaphase 1: Line up the homologous chromosomes along the equator of the spindle. Draw lines from the centrioles to the centromeres of each homologous pair to represent the spindle fibres.



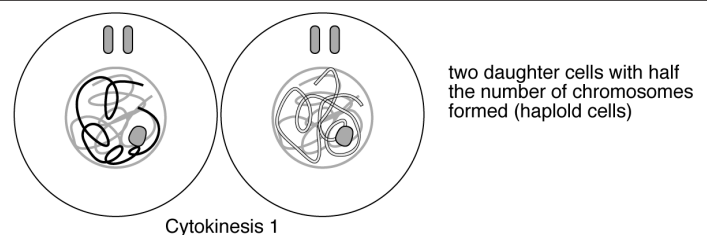
Anaphase 1

Anaphase 1: Do not separate the chromatids. Move the two stranded homologous chromosomes to opposite poles.



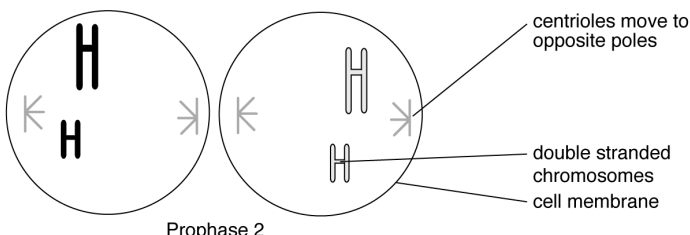
Telophase

Telophase 1: Show a constriction forming during telophase. Draw a nuclear membrane around the chromosomes. Draw 2 centrioles on top of the nucleus.



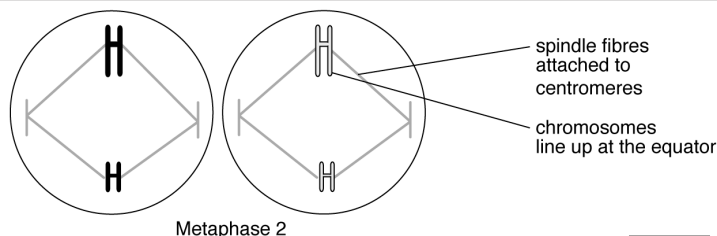
Cytokinesis 1

Cytokinesis 1: Draw 2 circles to represent 2 daughter cells. Place a long and short chromosome in the centre of the nucleus. Draw centrioles above the nucleus. Two daughter cells with half the number of the parent cells (2 chromosomes) are produced.



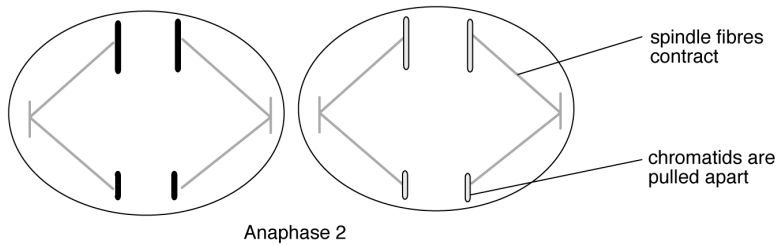
Prophase 2

Meiosis 2 – Prophase 2: Do not draw nuclear membrane and nucleolus. Draw centrioles with short spindle fibres at the opposite poles of each cell. Place 2 chromosomes in the middle of the circle.

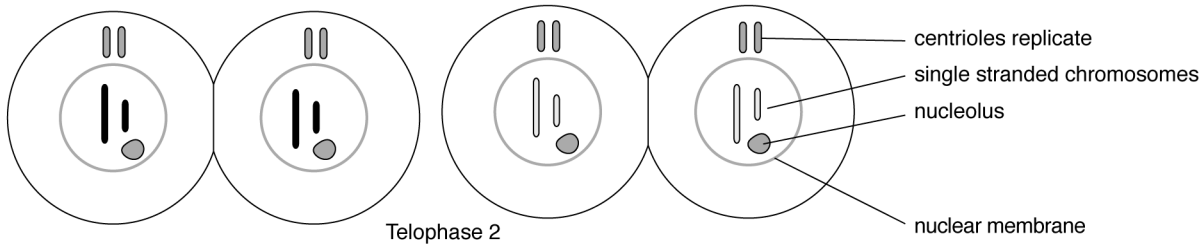


Metaphase 2

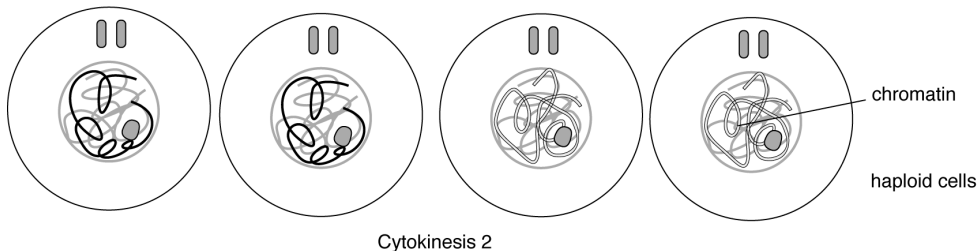
Metaphase 2: Line up the chromosomes at the cell's centre. Draw spindle fibres from the centrioles to the centromeres of each chromosome.



Anaphase 2: Draw shorter spindle fibres. Remove the pipe cleaners and move each chromatid to the opposite poles of the two cells.



Telophase 2: Draw a constriction forming in the middle of the cell. Draw nuclear membrane, nucleolus and centrioles. Place 2 single-stranded chromosomes in each daughter cell.



Cytokinesis 2: Draw 4 circles with nuclei and nucleoli. Draw 2 centrioles on top of the nucleolus of each daughter cell. Place two chromosomes in the nucleus.

Results

The results are the diagrams you drew.

Analysis of results

- 1 Name the cells that divide by mitosis and meiosis.
- 2 What is the diploid number of the parent cell in your model?
- 3 How many paternal and maternal chromosomes are present in the parent cell?
- 4 What is meant by homologous chromosomes?
- 5 At which stage in meiosis do homologous chromosomes pair?
- 6 At which stage does crossing over occur? Describe what happens during this process.
- 7 Why is meiosis known as reduction division?

Conclusion

- 1 Use the following points to draw up a table comparing mitosis and meiosis.

• Where and how often it occurs	• Anaphase
• Number of nuclear divisions	• Telophase
• Prophase	• Meiosis 2
• Metaphase	• Number of daughter cells at the end of meiosis 2
- 2 What is the advantage of meiosis?

ANTIBIOTIC RESISTANCE IN BACTERIA



Background

Timing: 4 × 45 minute lessons

Antibiotics are compounds that kill **bacteria**. The increase in **antibiotic-resistant bacteria** is largely due to the widespread use of antibiotics – in medicine, animal care and agriculture. *Staphylococcus albus* (*S. albus*) are spherical, gram-positive bacteria that take up the dye crystal violet and turn blue-violet in colour. They exist in clusters on our skin and have become resistant to multiple antibiotics. *Escherichia coli* (*E.coli*) are rod-shaped, gram-negative bacteria (do not take up the stain). These bacteria exist in our intestines and provide vitamin K and vitamin B complex. They are more susceptible to antibiotics. However, **E.coli O157:H7** can cause haemorrhaging (loss of blood).

The **Kirby-Bauer Method** can be used to test the antibiotic resistance of bacteria. In this method, an agar plate is covered with a **bacterial culture**. Paper disks containing the specific natural or synthetic antibiotics are then placed on the agar plate. After the agar plate is incubated, the **zone of inhibition** around each disk is observed and measured. The zone of inhibition is the clear area around the disk in which bacteria are not able to grow. If a bacterial colony is resistant to a particular natural antibiotic, then no zone of inhibition will form around the antibiotic disk.

SAFETY RULES

- Wear a lab coat and disposable gloves. Long hair should be tied back.
- Swab your bench space with a sponge and alcohol before and after completing the experiment.
- Do not remove test tube caps and petri dish lids because they can get contaminated with unwanted bacteria.
- Lift petri dish lids partially and replace as soon as possible. Place petri dishes in the incubator right side up. Never open a petri dish after you have inoculated it and have allowed it to incubate overnight.
- Discard all the petri dishes, cotton swabs and other disposables in autoclave bags after the experiment is finished. Do not put in the rubbish bin.
- Wash your hands before and after the experiment with a germicidal soap or alcohol.

- Equipment**
- distilled water
 - eucalyptus oil
 - tea tree oil
 - honey
 - nutrient agar medium (sterile) [peptone/beef poured in agar plates with lids]
 - filter papers
 - scissors
 - sterile swabs
 - 2 antibiotic multodisks
 - marking pen
 - cultures of *E.coli* and *S.albus*
 - forceps
 - alcohol
 - Bunsen burner
 - incubator
 - test tubes
 - cotton wool
 - 70% alcohol
 - metric ruler
 - adhesive tape

Part A – Testing the effect of natural antibiotics on *E. coli* and *S. albus*

Aim To investigate the effect of natural antibiotics on *E. coli* and *S. albus* by using the Kirby-Bauer Method.

Method

- 1 Clean your work area and sterilise it with 70% alcohol.
- 2 Use a marking pen to label 4 nutrient agar plates with your group number and the date. Mark two agar plates *S. albus* and the other two plates *E. coli*.
- 3 Remove the cap on the *S. albus* culture tube. Carefully flame the culture tube opening. Insert the sterile swab into the culture of *S. albus*. Squeeze the extra fluid from the swab by gently pressing the swab against the inner side of the culture tube. Remove the swab, flame the opening of your culture tube and recap it.

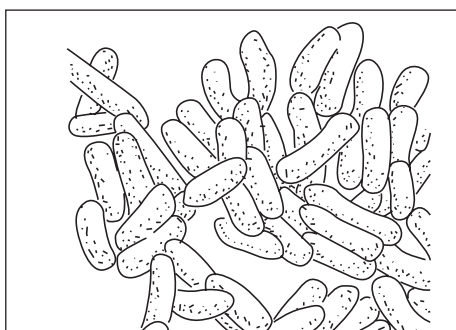
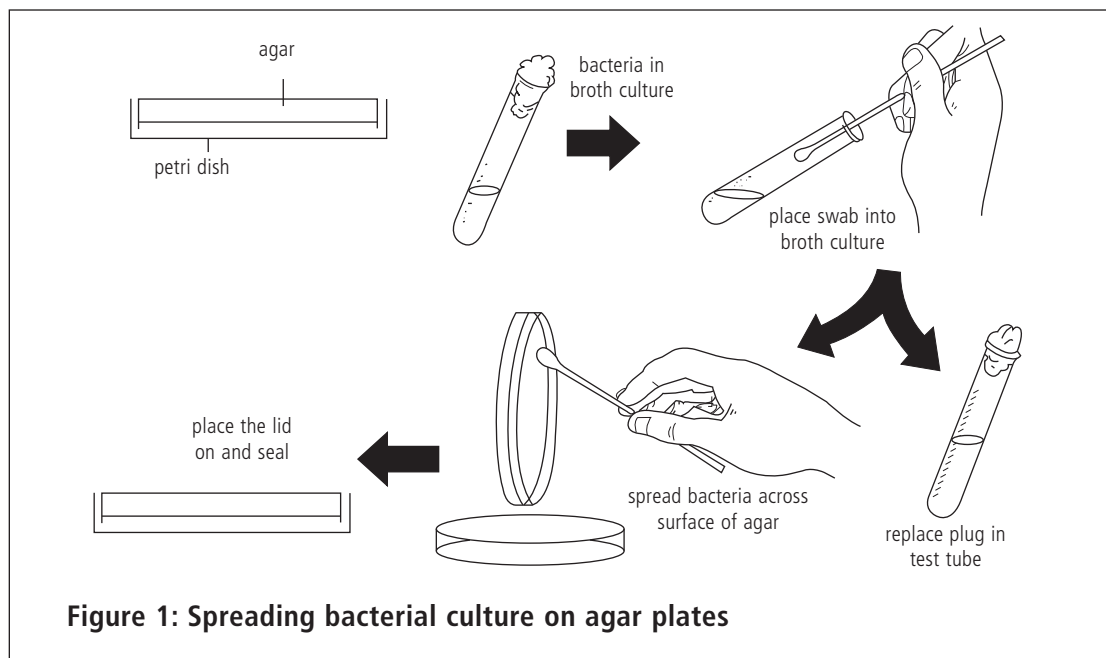


Figure 2: *E. coli* (rod-shaped bacteria)



Figure 3: *S. albus* (spherical clusters)

- 4 Use the swab to gently streak the entire agar surface of the two plates labelled *S. aureus*. Be sure to turn the plate at least three times while streaking to spread the bacteria evenly. Finally, gently run the swab around the edge of the agar plate. Cover your plate immediately. Discard the swab as directed by your teacher. Allow the covered plate and culture to dry for 5 to 10 minutes at room temperature.
- 5 Using a new sterile swab, repeat steps 3 and 4 to place the *E. coli* culture on the two plates marked *E. coli*.
- 6 Use one plate labelled *E. coli* and one plate labelled *S. albus* for part A. Use the other two plates for part B.
- 7 Divide the two plates into 4 sections. Label the four sections on the outside of the bottom part of each petri dish to indicate the antibiotics. D: distilled water, E: eucalyptus oil, T: tea tree oil, H: honey. (See figure 4.)
- 8 Cut 8 filter paper discs measuring 1 cm in diameter.
- 9 Sterilise the forceps by dipping it in alcohol and allowing it to air dry. Use the sterile forceps to pick up one disk dipped in distilled water and place it in the centre of section 1 on the *S. albus* plate.
- 10 Dip separate discs into eucalyptus, tea tree oil and honey. Place these on the agar surface in sections 2, 3 and 4. Make sure that the discs touch the agar but do not touch each other.
- 11 Repeat steps 9 and 10 with the dish infected with *E. coli*.

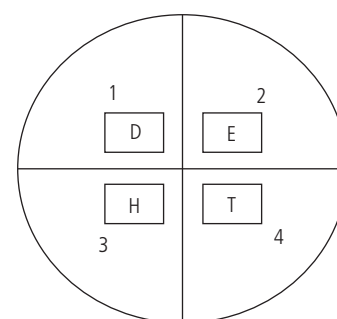


Figure 4: Placing natural antibiotics on agar plates

- 12 Seal the petri dishes with adhesive tape. Place them right side up in an incubator at 37°C for 3 days.
- 13 After 24 hours in the incubator, check the dishes for the presence of antibiotic activity by looking for a clear area, the **zone of inhibition**, surrounding the paper disc. **Caution:** Never open the dish for a better view.
- 14 After 3 days, determine the affectivity of each antibiotic by measuring the diameter of the zone of inhibition in mm. (The larger the diameter, the greater the resistance of the bacteria.)
- 15 Collate the class data. Then find the average zones of inhibition for each of the antibiotics before recording your data.
- 16 Determine which bacteria is susceptible (sensitive), unaffected (resistant) or (intermediate) for each of the antibiotics.

Results

- 1 Draw diagrams to show the zones of inhibition seen in the *E. coli* and *S. albus* dishes.

Table 1

Natural antibiotics	Zone of inhibition (mm)		
	<i>E. coli</i> Gram negative	<i>S. albus</i> Gram positive	Interpretation (Sensitive, intermediate or resistant)
1 Distilled water (D)			
2 Eucalyptus oil (E)			
3 Honey (H)			
4 Tea tree oil (T)			

Analysis of results

- 1 What was the purpose of using the disc dipped in distilled water?
- 2 What was the independent variable in this experiment? What was the dependent variable?
- 3 Which of the natural substances were effective antibiotics?
- 4 Did you notice any differences between your results and those of other groups in the class?
- 5 Describe the possible sources of errors in this experiment. How would you rectify these sources of errors?

Part B – Testing the effect of commercial antibiotics

Aim To investigate the effect of synthetically prepared antibiotics contained in multodiscs on *E. coli* and *S. albus* by using Kirby-Bauer method.

Method

- 1 Take the two agar plates prepared in part A, one infected with *E. coli* and the other with *S. albus*.
- 2 Use a sterilised forceps to pick up the multodisc and place it in the middle of *E. coli* dish. Make sure that each disc makes contact with the agar.
- 3 Repeat step 2 to place a multodisc in the middle of the *S. albus* dish.
- 4 Incubate the agar plates at 35°C for 24 hours.
- 5 The next day, observe the plates for zones of inhibition. Use a metric ruler to measure the zone of inhibition in millimetres. Measure across the zone and the antibiotic disk to find the diameter. Record the measurements in Table 2.

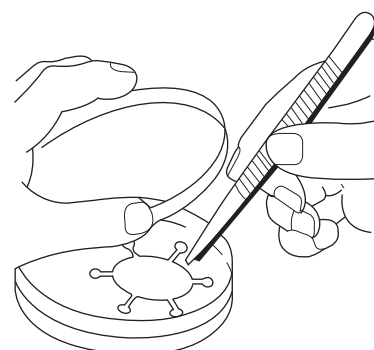


Figure 5: Placing the multodisc on agar

- 6 Use your observations and measurements to determine whether the bacteria are resistant or sensitive to each antibiotic. Collate the class results.
- 7 Put the petri dishes back into the incubator for another 24 hours to see if there is any change in the appearance of the zones of inhibition.

Note

- Do not touch the agar surface with the forceps or place the forceps on the lab table. If you do, you will need to sterilise them again.
- Do not press the antibiotic disks into the agar, but be sure that contact is made between the disks and the surface of the plate.
- Do not invert the agar plates when incubating. This will prevent the antibiotic disks from falling off overnight.
- Do not open the agar plates once they are inoculated with bacteria. Keep the plates closed when recording data.
- Take the measurements by flipping the plate over.
- Dispose of the agar plates

Results

- 1 Draw diagrams to show zones of inhibition in *E. coli* and *S. albus* plates.

Table 2

Antibiotics	Zone of inhibition (mm)		
	<i>E. coli</i> Gram negative	<i>S. albus</i> Gram positive	Interpretation (resistant or not resistant)
Ampicillin (A)			
Chloramphenicol (C)			
Penicillin (P)			
Streptomycin (St)			
Sulphatriad (S)			
Tetracycline (T)			

Analysis of results

Use the class data to answer the questions.

- 1 What is the advantage of using a multodisc instead of paper discs soaked in antibiotics?
- 2 Did you see any differences between *S. albus* and *E. coli* in their resistance to any of the antibiotics? If yes, state the differences.
- 3 Suggest why the zones of inhibition in two agar plates were different for different antibiotics.

Conclusion

- 1 What is an antibiotic?
- 2 What are the causes of antibiotic resistance in bacteria?
- 3 What conclusion can be drawn from the results of part A and part B?

Background

Timing: 2 × 45 minute lessons

Parasites are organisms that live in association with, and at the expense of, other organisms. The organism that provides the nutrition is called the **host**. The organism obtaining the nutrition is the **parasite**. Parasites that are found on the exterior of a host are called **ectoparasites**. Parasites found inside a host are called **endoparasites**. The effect of a parasite on a host ranges from minor harm (allowing the host to live and reproduce normally and complete its normal life cycle) to completely interfering with reproduction or causing the premature death of the host.

Definitive or **primary hosts** are those organisms on which a parasite reaches sexual maturity. **Secondary** or **intermediate hosts** are required by parasites to complete their life cycles. The parasite usually undergoes morphological or physiological changes while on the secondary host. A **vector** is a host that plays an active role in transmission. It can be a primary or a secondary host.

Most parasites are dorso-ventrally flattened. This allows them to move freely. They have organs of attachment for clinging on to their hosts, such as hooks, suckers and spines. Parasites usually have no nervous system, excretory systems or sense organs. If they live within the gut or blood of their hosts, they have a reduced gut, absorb nutrients through the whole body surface, and have a thick, resistant cuticle. They usually produce eggs and sperm throughout the year. They also reproduce by **asexual reproduction** at different stages of the life cycle to increase in number. By infecting more than one host species, the parasite can survive periods when one host is temporarily scarce by spreading out to different hosts and areas.

Aim

To observe the life cycles of a **tapeworm** and a **head louse** and the structural adaptations that enable them to survive and reproduce in their hosts.

Equipment • preserved specimens and life cycles of a tapeworm and a head louse

- stereo microscope
- compound light microscope
- microscope lamp
- head louse slide
- scolex slide
- proglottid slide

SAFETY RULES

- Be careful when handling the light microscope and the stereo microscope.
- Be careful when observing prepared slides.
- Use the low-power objective of the light microscope to observe the mouth parts, scolex and proglottid in detail, as the parasites you are observing are macroscopic.
- If you want to observe a whole organism, use the stereo microscope.
- Pack up the microscopes, slides and specimens as instructed by your teacher.

Part – A Study of endoparasites: Pig tape worm (*Taenia solium*) and a beef tape worm (*Taenia saginata*)

The tapeworms belong to the phylum **platyhelminthes** and the class **cestoda**. **Taenia solium** is the pig tapeworm. It uses humans as its primary host and pigs as its secondary host. **Taenia saginata** is the beef tapeworm. It uses humans as the primary host and cattle as the secondary host.

An adult tapeworm has a knoblike head, or **scolex**, equipped with **hooks** and **suckers** for attaching to the intestinal wall of the host; a **neck region**; and a series of flat, rectangular body segments, or **proglottids**, generated by the neck. The chain of proglottids may reach a length of 7 metres in *T. solium* and 12 metres in *T. saginata*. The adults have an average of 1000–2000 proglottids. Terminal proglottids break off and are excreted in the faeces of the host, but new ones are constantly formed at the anterior end near the neck. They have a rudimentary nervous system and excretory system and have no digestive tract. They absorb the host's digested food through their cuticle, or outer covering. Each proglottid contains a complete set of male and female reproductive organs (**hermaphrodite**) that produce the sex cells. Fertilisation is internal and about 50 000 eggs are produced per proglottid. The eggs are stored until the proglottid detaches from the tapeworm, migrates to the anus and is excreted in the stool. The eggs can survive for days to months in the environment.

Adult tapeworms rarely cause problems. In exceptional cases, the tapeworms might physically block the intestinal tract or proglottids might become lodged in the appendix and result in appendicitis. They do cause malnutrition and general debility. If humans consume water or food polluted with human faeces containing *T. solium* eggs, cysts can form in the muscle of the eye and cause **cysticercosis**. If the eggs move to the brain, they cause **neurocysticercosis** which can be fatal.

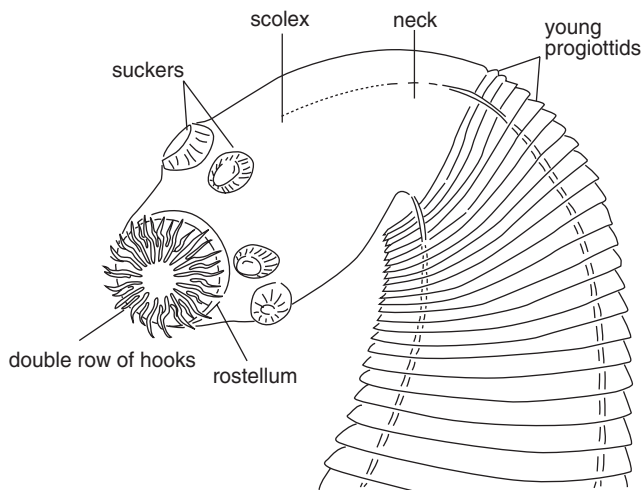


Figure 1: The scolex of a tapeworm

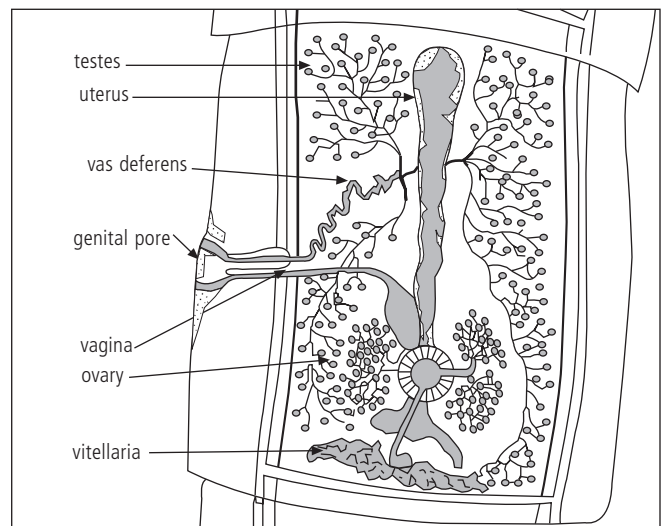


Figure 2: A mature proglottid of a tapeworm

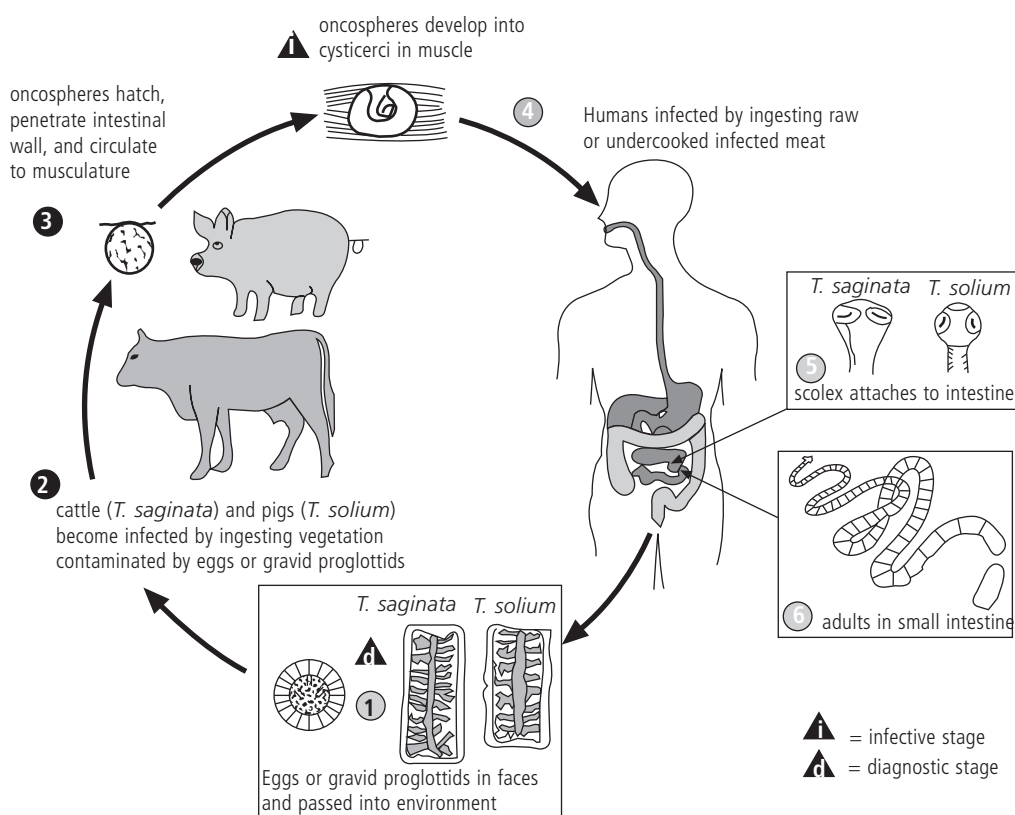


Figure 3: The life cycle of a pig tapeworm (*Taenia solium*) and a beef tapeworm (*Taenia saginata*)

Pigs or cattle become infected by eating vegetation that is contaminated with eggs or gravid proglottids. In the animal's intestine, the **oncospheres** hatch, penetrate the intestinal wall, and migrate to the striated muscles where they develop into **cysticerci** (bladder worms).

The bladder worm develops a small, fluid-filled sac called a bladder. This bladder surrounds the worm and provides nourishment. When the bladder is consumed by the definitive host, the head or scolex of the tapeworm is released. The scolex then attaches itself to the intestinal wall, grows, and begins to produce buds called **proglottids** at its posterior.

The proglottids remain attached to each other for a time. As they mature, each proglottid develops both male and female sex organs. The most mature proglottids eventually break loose, are passed out in the faeces and the life cycle continues.

Prevention of tapeworm infection

T. solium has only two hosts – pigs and humans. The only way to control cysticercosis is to break the cycle between the two. The use of latrines to ensure that pigs have no access to human excrement, washing hands before eating and thoroughly cooking pig meat are relatively simple measures. Vaccines that provide 93 to 95% protection for pigs and cattle are available.

Method

- 1 Use a compound light microscope to examine the scolex and proglottid slides on low-power objective. Note the hook-like structures and suckers on the scolex.
- 2 Examine the diagram of the life cycle of the tapeworm. Note the various stages it goes through in primary and secondary hosts.

Results

- 1 Draw diagrams of a scolex and a proglottid of the tape worm.

Analysis of results

- 1 Name the primary and secondary hosts of beef tapeworms and pig tapeworms.
- 2 Describe the structure and function of the scolex and the proglottid.
- 3 How does the tapeworm get its nourishment?
- 4 How do humans become infected with bladder worms?
- 5 What is the advantage of reproducing by asexual reproduction and having two hosts?
- 6 What problems do these tapeworms cause their hosts?
- 7 Find out about the methods used to prevent and control the transmission of these tapeworms from host to host.
- 8 Complete the table.

Stage in the life cycle	Where it occurs	Role played by this stage
Adult tapeworm		
Eggs		
Oncosphere		
Cysticircus (bladder worm)		

Part B – Study of an ectoparasite: Head lice (*Pediculus humanus capitus*)

Head lice are **ectoparasites**. They belong to the phylum **arthropoda** and class **insecta**. They are small (3 mm in length), wingless insects with sucking mouthparts and 3 pairs of legs designed for grasping hairs. Their body is dorsoventrally flattened, which allows them to move freely in the hair. Head lice obtain nutrients by sucking blood from the host's scalp. They cannot fly or jump and do not travel far from the host.

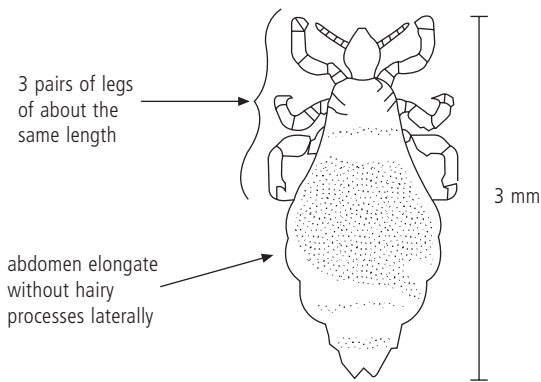


Figure 4: Structure of a head louse

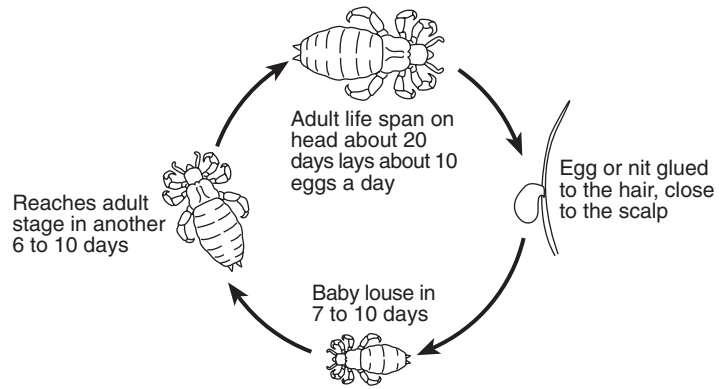


Figure 5: Life cycle of a head louse

The lifecycle of a head louse

Head lice are **unisexual**. Mated females deposit eggs daily over a period of 20–30 days. **Eggs (nits)** are cemented to the hairs of the host. The grayish-white nits (1 mm in length) are shaped like an elongated football. They have a cap at one end to admit air and allow the young insect to escape and they hatch in 8 to 10 days. The **nympal louse** (immature lice) resemble the adults except for their size. They feed on blood and moult three times to allow for body growth before becoming sexually mature. An adult louse can live for up to 40 days.

Head lice are mainly acquired by direct head-to-head contact with the hair of an infested person. They can also be transferred through shared combs, hats and other hair accessories. They rarely cause direct harm and do not transmit infectious agents among people. The presence of head lice may cause itching, loss of sleep and allergic reaction in some people. The louse's saliva and faeces may sensitise people to their bites, which increases the irritation and the chance of secondary infections due to excessive scratching.

Prevention of head lice

Children should be encouraged not to share combs, hats and other personal belongings. Daily washing, the changing of clothes and keeping the hair short will also help discourage lice. Periodic inspections aid in early detection. A good louse or nit comb must be used to locate and remove the insects and their eggs. Shampoos containing permethrin or pyrethrins can be used to kill the insects.

Method

- 1 Use a stereo microscope to observe a preserved specimen of a head louse. Note its size, the shape of the body, the number of legs and the types of mouth parts.
- 2 Look at the life cycle diagram. Note the stages head lice go through and the time taken for each stage.

Results

- 1 Draw and label a head louse and its life cycle.

Analysis of results

- 1 Describe the size and shape of the head louse.
- 2 How are the mouth parts adapted for getting nutrients from its host?
- 3 What are the stages the head louse goes through during its life cycle?
- 4 How many hosts are involved during its life cycle?
- 5 How are head lice transmitted? What problems do they cause their hosts?
- 6 Find out about the methods used to prevent and control the transmission of head lice from one host to another.

Conclusion

Describe the structural adaptations of tapeworms and head lice that allow them to live and reproduce successfully in/on their hosts.



Homeostatic control of temperature after exercise

Results

Table 1 Individual group results

Note: Results will vary

	Rest	Exercise	Recovery					
Time (minutes)	0	2	—	0	2	4	6	8
Pulse rate per minute	About 72 bpm		—	Increases to 100 + bpm				Returns to resting rate
Breathing rate per minute	15–20 per minute		—	Increases to 60–80 per minute				Returns to resting rate
Carbon dioxide output (drops of NaOH)	About 5–10 drops		—	20–30 drops (can vary)				About 30 drops
Temperature (°C)	36°C–37°C		—	37°C–38°C				36°C–37°C

Percentage increase in heart rate: Results will vary. Generally in the range 40%–140%.
Change in cobalt chloride paper: Before exercise: no change, After exercise: changes to pink.
Skin colour: Before exercise: not flushed, After exercise: flushed or pink.

Table 2 Class results

Results will vary.

Table 3 Class results: percentage increase in heart rate

Results will vary. Generally in the range 40%–140%.

Analysis of results

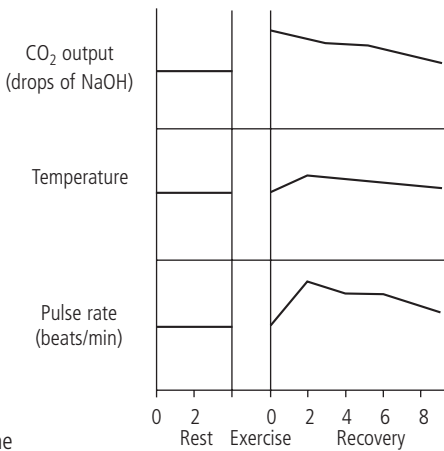
1 The subject's temperature increased immediately after exercise. **2** The temperature stayed constant with minor fluctuations. **3** The pulse rate increased after exercise. **4** The pulse rate increased after exercise because the heart beats faster after exercise to supply more blood to the muscles. **5** The breathing rate increased after exercise. **6** As the heart pumped the blood faster, more glucose and O₂ were supplied to the muscles and more CO₂ and wastes were removed. The breathing rate increased at the same time to get rid of the excess CO₂ and to supply more O₂. **7** CO₂ was removed from the leg muscles and O₂ was supplied. **8** The blood carries glucose to the muscles to supply energy. **9** The blood removed CO₂ and waste materials from the muscles. **10** The percentage increase in heart rate generally varies considerably between individuals. **11** The rate of heartbeat can be influenced by the age, sex, and fitness of the person. **12** The subject did sweat after exercise. As sweat evaporates from the skin, the heat from the skin is lost. This produces a cooling effect. **13** The skin became flushed (pink in colour). **14** The blood vessels near the skin dilate after exercise so that more blood can be brought to the skin. Heat is lost from the skin by conduction, convection and radiation. **15** Temperature receptors in the hypothalamus detect changes in temperature. Changes in pH (or CO₂ in the blood) are detected by osmo-receptors in the brain. **16** Suggestions for improvement. **a** Use a digital thermometer which is easy to read. **b** Measure the pulse rate with a stethoscope. **c** Use a respirometer to measure the breathing rate. **d** Use an exercise bike or a treadmill for exercising instead of running in order to regulate speed and distance. **e** Use freshly made NaOH throughout the experiment to eliminate variability in colour changes.

Conclusion

Temperature homeostasis is controlled by the thermoregulatory centre in the hypothalamus. It receives input from two sets of thermoreceptors. **1:** Receptors in the hypothalamus itself monitor the temperature of the blood as it passes through the brain (the core temperature). **2:** Receptors in the skin monitor the external temperature.
When the temperature increases after exercise, the thermoregulatory centre sends impulses to several different effectors to adjust the body temperature. Smooth muscles relax, causing vasodilation. More heat is carried from the core to the surface where it is lost by convection and radiation. The skin turns red. Sweat glands secrete sweat onto the surface of the skin, where it evaporates and produces a cooling effect. Adrenal and thyroid glands stop releasing adrenaline and thyroxine and the metabolic rate is reduced. The response is a decrease in temperature until it returns to normal.

Graphing

Expected shapes of graphs:





Phototropism and geotropism in plants

Part A

Hypothesis: Wheat seedlings with aluminium caps will not bend towards light, but seedlings without caps and seedlings with collars will bend towards light.

Results

Table 1

Experimental set up	Response to light	
	Treated seedlings	Untreated seedlings
Set up A: coleoptile tips cut	Did not bend	Bent towards light
Set up B: coleoptile tips covered with caps	Did not bend	Bent towards light
Set up C: coleoptiles with collars	Bent towards light	Bent towards light

Analysis of results

1 Untreated wheat seedlings were used as controls in each dish. 2 Wheat seedlings without caps and seedlings with collars bent towards the light because the tips were exposed to light. 3 The wheat seedlings exhibited positive phototropism. 4 Yes, the findings did support the hypothesis. (Answers will vary according to the hypothesis.) 5 Because the shoots grow towards the light, plants can photosynthesis and produce carbohydrates.

Part B

Hypothesis: Wheat seedlings will bend more under red and blue light than under yellow and green light.

Results

Table 2

Colour filters	Wavelength of light (nanometres)	Angle of bending
		24 hours
Clear	400–700 nm	More bending
Red	650–700 nm	More bending
Yellow	560–600 nm	Very little bending
Green	500–560 nm	No bending
Blue	430–500 nm	More bending

Analysis of results

1 The clear filter was used as control to compare the angle of bending that occurred under filters of different colours. 2 The seedlings bent the most under red and blue light. This is because the chlorophyll absorbs red and blue light and reflects green light. 3 The findings were supported by the hypothesis. (Answers will vary according to hypothesis.)

Part C

Hypothesis: No matter what position the bean seeds are planted in, roots will always grow towards gravity, exhibiting positive geotropism.

Analysis of results

1 Direction of root growth. 12 o'clock: Roots will grow downwards. 9 o'clock: Roots will grow and bend downwards, then continue growing downwards. 6 o'clock: Roots will grow upwards then bend and grow downwards. 3 o'clock: Roots will grow and bend downwards, then continue growing downwards. 2 All the roots exhibited positive geotropism. 3 By growing towards the ground, plant roots can obtain water and nutrients from the soil.

Conclusion

1 The bending of shoots towards light is known as positive phototropism. The bending of roots downwards according to gravity is known as geotropism. 2 Auxin produced by the shoot tips goes to the dark side of the shoot and promotes the elongation of cells on that side. This causes the bending of the shoot towards light. Auxin concentrations high enough to stimulate stem growth inhibit root growth. When a root is placed on its side, amyloplasts settle by gravity to the bottoms of cells in the root tip. This inhibits root cell elongation. So the cells at the top surface of the root elongate, causing the root to grow down. 3 Plants bend more towards red and blue light than towards green and yellow lights because the chlorophyll in leaves absorbs red and blue light and reflect green light. 4 The position of the seed did not affect the growth of the root. The roots always grew towards the ground/gravity.



Investigating reflex actions

Part A – Measuring the speed of a nerve impulse

Results

Table 1: Measurements will vary. Generally, with the eyes open, reaction times will be faster for the dominant hand than for the non-dominant hand. All of the eyes closed reaction times will be slower than the eyes open reaction times. Of the eyes closed reaction times, the tap on the hand will generally be a shorter reaction time than the tap on the hand or foot.

Analysis of results

1 The reaction time to catch the ruler decreased over the 4 trials. **2** The dominant hand reaction time is faster than the non-dominant hand. **3** For the experiments where the subject's eyes were closed, when head was tapped the reaction time was fastest, when the hand was tapped was next fastest, when the foot was tapped was slowest. **4** This type of reaction is voluntary, because the action is under conscious control.

Part B – Investigating reflex actions

Results

Table 2

Reflex measured	Observation	Interpretation
Blink reflex	The eyes blinked when the cotton ball was thrown.	This happens to protect the eyes from harm.
Pupillary action	Compared to the pupil diameter in low light, the pupil contracted when the torch was shined in the eye.	This happens to control the amount of light entering the eye.
Knee reflex	The knee jerked when it was hit by the rubber hammer.	The knee-jerk reflex is a falling reflex that, when we are off balance, 'kicks' us into the air to help us recover our footing.

Analysis of results

1 The subject blinked when the cotton ball was thrown at the eyes to protect their eyes. **2** The size of the pupil became smaller in light and larger in the dark to control the amount of light entering the eye. **3** When the subject's knee was tapped, the leg jerked forwards. **4** The message is sent from the sense organs to the brain via sensory nerves. **5** The brain sends the message to the muscles via motor nerves. **6** Interneurons are found in the spinal cord, which connects the sensory nerves to the motor nerves.

Conclusion

1 The subject reacted faster with their eyes open because the brain received the message faster from the eyes. When the eyes were closed, they had to get the message from the area of the skin which was tapped. This takes longer to reach the brain. **2** A reflex action is not under voluntary control. It is controlled by the midbrain and the spinal cord. **3** A reflex action protects a person from immediate danger. **4** Reflex testing is an important diagnostic tool for assessing the condition of the nervous system. Distorted, exaggerated, or absent reflex responses may indicate degeneration or pathology of portions of the nervous system, often before other signs are apparent. If the spinal cord is damaged, then reflex tests can help determine the area of injury.



Sense receptors in the skin

Part A – Investigation of heat and cold receptors

Results

Table 1 Sensations felt by forefingers in water

Finger used	Sensation felt: hot, warm or cold		
	Cold water	Water at room temperature	Warm water
Right forefinger	cold	warm	
Left forefinger		cold	warm

Analysis of results

1 (Note: Answers could vary.) When the subject placed the right forefinger in cold water it felt cold, but when it was placed in water at room temperature it felt warm. **2** (Note: Answers could vary.) When the subject placed the left finger in warm water it felt warm, but when it was placed in water at room temperature it felt cold. **3** When temperature receptors get used to a certain amount of heat or cold, they adapt and become less sensitive.

Part B – Investigation of touch receptors

Results

1 Subject's perception about the weight of the coins: generally, the coins will feel the same weight.

Table 2: The actual two-point distances will vary. Generally, the finger tip and upper lip are very sensitive; the cheek, nose, palm and forehead are sensitive; and the forearm and upper arm are mildly sensitive.

4 Sense receptors in the skin (cont'd)

Analysis of results

1 Results will vary between groups due to differences in individual physiology; the use of slightly different probes; and variations in the accuracy of measurements. **2** Fingertips and lips are best at recognising two points even when the points are very close together. **3** Areas with smaller two-point distances have more receptors. **4** The area of the brain receiving information from skin that has many receptors will be larger. **5** The information from sensory receptors in the skin is sent to the brain via sensory nerves. **6** Make sure regular distances between the toothpicks are used, such as 40 mm, 35 mm, 30 mm. Repeat the experiment several times on the same subject and calculate an average measurement. Use several different corks that already have the toothpicks at specific distances to reduce errors in measurement.

Conclusion

1 Touch and pressure receptors detect the weight of the coins. Heat and cold receptors detect the temperature of water. Touch receptors detect two-point discrimination. **2** Lips and fingertips are most sensitive to two-point discrimination. **3** Touch/pressure receptors: Unevenly distributed in the epidermis and dermis of the skin. Pain receptors: Throughout the dermis region of the skin. Heat and cold receptors: Throughout the dermis region of the skin.



5 Orientation movements exhibited by planaria

Part A – Reaction to light and touch: phototaxis and thigmotaxis

Hypothesis: Planaria will move away from light and exhibit negative phototaxis. If the planarian is touched it will contract and exhibit thigmotaxis.

Results

Table 1 Reaction to light

Observation	Dish A	Dish B	
	Lighted constantly	Area 1 lighted	Area 2 lighted
Number of planaria	10	0	0
Movement of planaria (speed and direction)	Move around in the lighted area	Move away from lighted area towards shade	Move away from lighted area towards shade

Reaction to touch: When a planarian is touched with a pipette it **contracts** and exhibits **thigmotaxis**.

Part B – Reaction to gravity: geotaxis

Hypothesis: Planaria will move towards gravity and exhibit positive geotaxis.

Results

Table 2 Planarian reaction to gravity

Time spent in the bottom part of the test tube (mins)	Time spent at top part of the test tube (mins)	Time spent at the middle of the test tube (mins)	Geotaxis: positive or negative
more	less	less	positive geotaxis

Part C – Reaction to chemicals: chemotaxis

Hypothesis: Planaria will move away from chemicals and exhibit negative chemotaxis.

Table 3 Reaction to chemicals

Time spent at the centre line (mins)	Time spent near vinegar (mins)	Time spent away from vinegar (mins)	Time spent near salt solution (mins)	Time spent away from salt solution (mins)	Chemotaxis: positive or negative
less	less	more	less	more	negative chemotaxis

Analysis of results

1 Dish A, which was constantly exposed to light, was used as the control in part A. **2** Dish B was partly covered by cardboard to test whether planaria move towards light or away from light. **3** When a planarian was touched with a pipette, it contracted and exhibited thigmotaxis. **4 Light:** When planaria were exposed to light they moved away from it, exhibiting negative phototaxis. **Gravity:** When planaria were exposed to gravity, they moved towards gravity, exhibiting positive geotaxis. **Vinegar and salt:** When planaria were exposed to vinegar and salt they moved away from them, exhibiting negative chemotaxis. **5** Planaria react positively towards gravity. Planaria react negatively towards light and chemicals. **6** Collating class results gives more reliable results. **7** Answers will vary, but generally when salt and vinegar are added the planaria die. **8** Observe 10 planaria for each type of behaviour; repeat the experiment several times; use more dilute solutions of salt and vinegar; observe the planaria under natural conditions.

Conclusion

Planaria exhibit negative phototaxis, positive thigmotaxis, positive geotaxis and negative chemotaxis movements when exposed to light, touch, gravity and chemicals.



Social and aggressive behaviour in fish

Results

Part A – The tendency of fish to school with their own species

Table 1: On average, both the zebra fish and the goldfish will spend more time with the same species and less time in the middle and with different species.

Part B – Shoal number preference

Table 2: On average, the zebra fish will spend more time with the group of 6 fish and less time in the middle and with the group of 2 fish.

Part C – The social and aggressive behaviour of catfish and Siamese fighting fish

Table 3

Behaviour pattern	Catfish with catfish	Siamese fighting fish with Siamese fighting fish	Catfish with Siamese fighting fish
Position of fins	Fin position did not change	Fins were spread out like wings, attacked the divider	Cat fish: did not change Siamese: spread its fin slightly
Position of tail	Did not change	Tail was spread out to look bigger	Did not change
Movement of mouth	Did not open and close frequently	Mouth opened and closed frequently	Did not open and close frequently
Movement of gill covers	Did not move much	Gill covers move up and down frequently	Did not move much
Colour change	No colour change	Colour becomes more brilliant	No colour change

Analysis of results

1 The test zebra fish spent more time near the jar with zebra fish. **2** The test zebra fish showed a preference for the larger shoal. **3** The test zebra fish showed a preference towards its own species. **4** Zebra fish, goldfish and catfish exhibited more social behaviour towards their own species and other species. Male Siamese fighting fish showed more aggressive behaviour towards another male Siamese fighting fish. **5** Because male Siamese fighting fish are more aggressive towards other male Siamese fighting fish. Female Siamese fish are not aggressive. **6** The male Siamese fighting fish recognise each other by colour and the display of fins and tail.

Conclusion

1 Zebra fish, goldfish and catfish are more social towards their own species. They are not aggressive to either their own species or other species. **2** Fish prefer to live in groups because they can defend against competitors and predators, have greater protection from predators, greater chance of finding food, greater efficiency in capturing prey when foraging. However the disadvantages of social living are the greater chance of being detected by a predator, greater food competition within the group and the greater chance of disease transmission. **3** Male Siamese fighting fish are more aggressive towards other male Siamese fighting fish especially during their breeding period. They can fight to the death for the privilege of mating with females.



Modelling mitosis

Part A – Examining a prepared slide of an onion root tip

Results

Table 1 and histogram: Results will vary.

Analysis of results

1 Onion root tips are chosen for the study of mitosis because the cells divide actively in this region, so all the different stages of mitosis are easily visible. **2** Cells are undergoing mitosis in the zone of cell division. **3** The root cap protects the root tip from the soil grains. In the zone of elongation, the cells elongate and become larger after cell division. In the zone of cell differentiation, the cells become differentiated into different types of cells. In the zone of maturation, the cells become mature and carry on various functions. **4** Answers will vary, but generally, more cells will be at the interphase stage. **5** The main difficulty is determining whether a cell is in interphase or prophase. In some cases, the nuclear membrane is still intact, but the nucleoli has already disappeared.

Part B – Modelling stages in Mitosis

Analysis of results

1 Karyokinesis is the division of the DNA in the nucleus. Cytokinesis is the division of cytoplasm. **2** Somatic cells divide by mitosis to replace old cells, to repair damaged cells and for growth. **3** Chromosomes align on the equator of the cell during metaphase. **4** There are 4 double-stranded chromosomes during the prophase stage. **5** There are 4 single-stranded chromosomes in the nucleus of each daughter cell.

Conclusion

Interphase: This is the resting stage of the cell. Chromosomes are coiled like a ball of wool and are called **chromatin**. The nucleolus, nuclear membrane and 2 centrioles are visible.

Prophase: The chromosomes untangle, shorten and start to replicate (2 copies are made). Each chromosome has duplicated and now consists of two sister chromatids. The nucleolus and nuclear membrane disappear. Each centriole moves to the opposite sides of the cell.

Metaphase: Chromosomes arrange themselves at the equatorial plane of the cell, one below another. Nuclear spindles form from the centrioles. Each chromosome is attached to the spindle at the centromere. All the chromosomes line up in the middle of the cell, one below another.

7 Modelling mitosis (cont'd)

Anaphase: The spindle fibres contract and pull the chromatids of the chromosomes apart. Each chromatid from the chromosome moves to opposite poles of the cell. The cell membrane starts to constrict and divide the cell into two daughter cells.

Telophase: The nucleolus and nuclear membranes appear. Two daughter cells are formed, each with the same number of chromosomes as the parent cell.

Cytokinesis: After karyokinesis, the cytoplasm of the cell divides. In animal cells, the cell pinches in the middle and two daughter cells are formed. In plant cells, a cell plate forms in the middle. Each daughter cell has the same number of chromosomes as the parent cell.



Modelling meiosis

Analysis of results

1 Somatic cells divide by mitosis and sex cells divide by meiosis. **2** The diploid number of the parent cells is 4 chromosomes. **3** The parent cell has 2 paternal and 2 maternal chromosomes. **4** One member of each pair of homologous chromosomes comes from the mother and the other from the father; they are similar in size and genetic constitution. **5** Homologous chromosomes pair during prophase 1. **6** When chromosomes pair in the early prophase 1 of meiosis 1, a cross over can occur between two non-sister chromatids of homologous chromosomes. This results in an exchange of genetic material between the maternal and paternal chromosomes, and can result in an exchange of alleles between chromosomes. Crossing-over produces genetic variation, the raw material for evolution. **7** Meiosis is known as reduction division because the 4 daughter cells that form at the end of meiosis are haploid and contain half the chromosome number.

Conclusion

Stages	Mitosis	Meiosis
Where and how often it occurs	Mitosis occurs frequently because it allows growth and repair of body tissues.	Meiosis only occurs at certain times in the life cycle of sexually reproducing organisms.
Number of nuclear divisions	There is only one nuclear division.	There are two nuclear divisions.
Prophase	There is no crossing over in mitotic prophase.	There is crossing over between adjacent homologous chromosomes in prophase 2 of meiosis.
Metaphase	Duplicated chromosomes align on the metaphase plate in single file.	Homologous chromosomes align in pairs on the metaphase 1 plate.
Anaphase	Sister chromatids separate to form daughter chromosomes.	Sister chromatids do not separate in anaphase 1 of meiosis. They separate in anaphase 2.
Telophase	2 diploid daughter cells that are genetically identical to the parent cell are formed.	2 haploid cells are formed at the end of meiosis 1.
Meiosis 2	Does not occur	Similar to mitotic division
Number of daughter cells at the end of meiosis 2	None produced	4 haploid daughter cells which are not genetically identical to the parent cell are formed.

2 Meiosis provides genetic variation which gives organisms the ability to adapt to changing environments. (In humans with 23 haploid chromosomes, $2^n = 2^{23} = 8,388,608$ combinations are possible. If only one cross over occurs within each homologous chromosome, 4^{23} or 70,368,744,000,000 combinations are possible.)



Antibiotic resistance in bacteria

Part A – Testing the effect of natural antibiotics on *E. coli* and *S. albus*

Results

Table 1 (answers may vary)

Natural antibiotics	Zone of inhibition (mm)		
	<i>E. coli</i> Gram negative	<i>S. albus</i> Gram positive	Interpretation (sensitive, intermediate or resistant)
1 Distilled water (D)	No clear patch	No clear patch	No bacteria were killed.
2 Eucalyptus oil (E)	Big clear patch	Big clear patch	More bacteria killed
3 Honey (H)	Smaller patch	Smaller patch	Some bacteria were killed.
4 Tea tree oil (T)	Big clear patch	Big clear patch	More bacteria killed

9 Antibiotic resistance in bacteria (cont'd)

Analysis of results

1 The disc dipped in distilled water was used as a control to compare with other discs. **2** Dipping the discs in different natural antibiotics is the independent variable. The resistance/ sensitivity of bacteria (the clear space around the discs) is the dependent variable. **3** Eucalyptus oil and tea tree oil were more effective natural antibiotics. **5** The amount of antibiotics used in each disc may have varied. There could have been some errors in measuring the clear zones. Measure the amount of antibiotic and place the discs in the correct place without any overlapping. Take the measurements carefully.

Part B – Testing the effect of commercial antibiotics

Results

Table 2: Answers may vary depending on results. *E. coli* is less resistant to most of the antibiotics except Penicillin. *S. albus* is resistant to sulphatriad and tetracyclin.

Analysis of results

1 All antibiotics in multodiscs are of the same concentration and do not overlap with each other. But if paper discs soaked in antibiotics are used, the concentration of the antibiotics cannot be controlled and they may diffuse across and cause experimental errors. **2** *S. albus* was resistant to sulphatriad and tetramycin. *E. coli* was susceptible to all antibiotics tested except for penicillin. (Answers will vary depending on results.) **3** The concentration of the antibiotic on the disks, the type of bacteria, the temperature and length of incubation, the thickness of the agar in the plates, the age and density of the culture, the age of the antibiotic disks and the laboratory technique used affect the zone of inhibition.)

Conclusion

1 An antibiotic prevents the growth of bacteria. **2** Unnecessary use of antibiotics by humans, use in animal feeds in low doses, high availability in many countries, misuse by health professionals and patient failure to follow prescribed courses of treatment. **3** Tea tree oil and eucalyptus oil are more effective natural antibiotics. Different antibiotics in multodiscs had different effects on *S. albus* and *E. coli*. *S. albus* was resistant to more antibiotics than *E. coli*.



Human parasites

Part A – Study of endoparasites: Pig tape worm and a beef tape worm

Analysis of results

1 Humans are the primary hosts and cattle or pigs are the secondary hosts. **2** The scolex has rows of hooks and suckers by which the tapeworm attaches itself to the walls of the small intestine. Each proglottid has a male and female reproductive system and has no nervous, digestive or excretory systems. It is covered by a thick cuticle. **3** It gets its nourishment by absorbing digested food through its cuticle from the intestines of the human host. **4** When humans eat undercooked pork or beef, they can get infected with bladder worms. **5** Asexual reproduction helps them to increase their number. By infecting more than one host species, the parasite can survive periods when one host is temporarily scarce and can spread out to different areas. **6** Tapeworms might physically block the intestinal tract or proglottids might become lodged in the appendix and result in appendicitis. They also cause malnutrition and general debility. If humans consume water or food polluted with human faeces containing *T. solium* eggs, cysts can form in the muscle, the eye or the brain, causing neurocysticercosis which can be fatal. **7** The cycle between the two hosts must be broken. The use of latrines to ensure pigs and cattle have no access to human excrement, washing hands before eating and thoroughly cooking meat are relatively simple measures. Vaccines are available to provide 93 to 95% protection for pigs and cattle.

Stage in the life cycle	Where it occurs	Role played by this stage
Adult tapeworm	Found in the small intestines of human hosts	Eggs are produced by sexual reproduction
Eggs	Excreted in the faeces and eaten by cattle or pigs	Moves from human host to secondary host: pig or cow
Oncosphere	Found in the intestine and in the muscle of the secondary host	Moves from the intestines to the muscle
Cysticercus (bladder worm)	Found in the muscles of the secondary host	Remains in the muscle until consumed by the human host

Part B – Study of an ectoparasite: Head lice (*Pediculus humanus capitus*)

Analysis of results

1 Head lice are small (3 mm long), wingless insects with sucking mouthparts, a head narrower than the thorax and 3 pairs of legs designed for grasping hairs. **2** Head lice have sucking mouthparts to suck blood from the host's scalp. **3** Stages in the life cycle of a head louse: egg, 1st nymph, 2nd nymph, 3rd nymph and adult. **4** Only one host (human) is involved in its life cycle. **5** Head lice are mainly acquired by direct head-to-head contact with the hair of an infested person, but can also be transferred through shared combs, hats and other hair accessories. **6** Children should be encouraged not to share combs, hats and other personal belongings. Daily washing, changing of clothes and keeping hair short will also help discourage lice. Periodic inspections aid in early detection. A good louse or nit comb must be used to locate and remove the insects. Shampoos containing pyrethrin or pyrethrins can be used to kill the insects.

Conclusion

Adaptations of a tapeworm: It has a thick cuticle which protects it from human digestive enzymes. It is a hermaphrodite or bisexual so it does need to search for a mate. Each proglottid produces numerous eggs. The scolex has hooks and suckers so that it can cling to the intestinal walls of the host.

Adaptations of a head louse: They have sucking mouth parts to suck blood. They have 3 pairs of legs to grasp the hair of their hosts. Their body is dorsoventrally flattened to enable them to move freely in the hair.

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